

Horizons of Truth
Gödel *Centenary* 2006



Gödel's Ontological Dream

Mathematical Realism, Temporal Unrealism, and Gödel's God



Professor Gary Mar

Gary.Mar@stonybrook.edu

Stony Brook University

Stony Brook, New York, USA 11794

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[Gödel Freundkreis](#) Essay Competition

Bamberg University, Germany

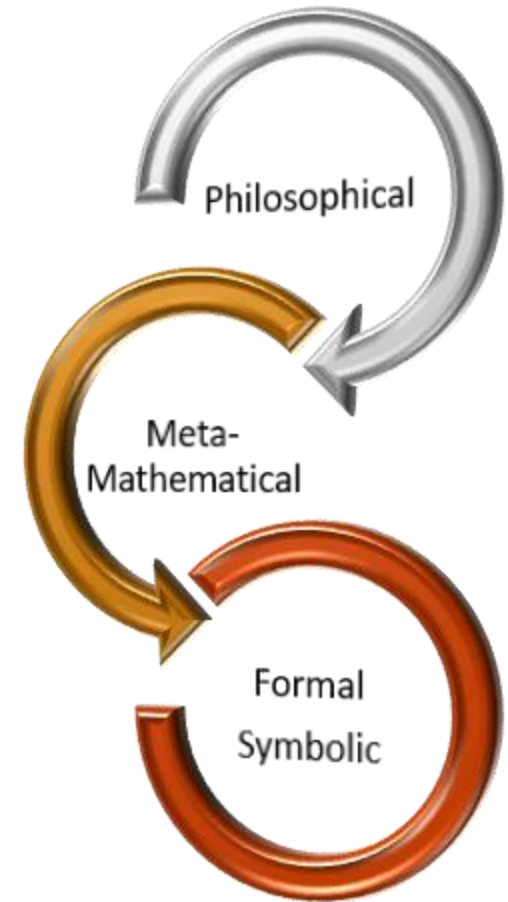
“Is all that we see or seem But a dream within a dream?”

Gödel’s theorems evoke the feeling of a “dream within a dream *within a dream*”

Hilbert’s enchantment effective procedures

Newton’s postulation of absolute space and time

Vienna Circle’s verificationism and atheism.



“The axiomatic method goes step by step”

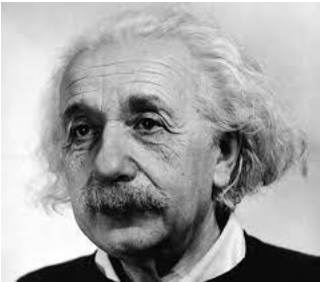
1. *A dream within a dream ...*
2. MATHEMATICAL PLATONISM
3. INCOMPLETENESS THEOREMS
4. MINDS VS. TURING MACHINES?
5. UNREALITY OF RELATIVISTIC TIME
6. GÖDEL'S ONTOLOGICAL DISCOVERY
7. *Awakening from a dream?*



*“Mathematicians are platonists during the week
but formalists on Sundays”*



G. H. Hardy, 4th Gibbs
Lecture [1928]



Einstein, 11th Gibbs
Lecture [1934]



27th Gibbs Lecture [1951]

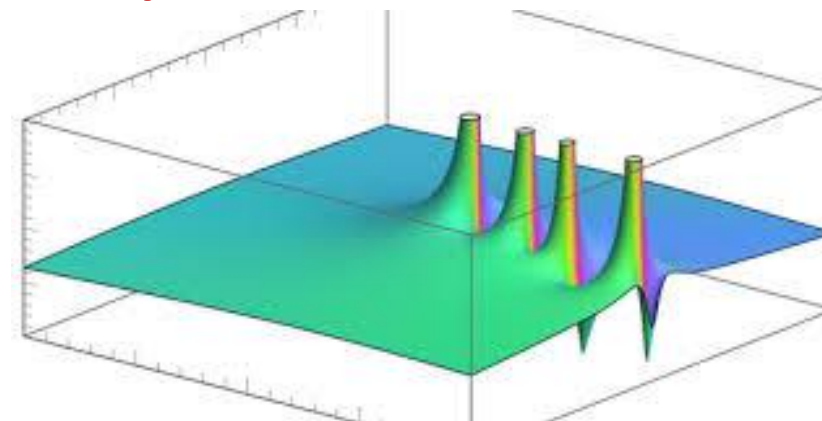
“The real argument for objectivism is the following. We know many general propositions about natural numbers to be true (2 plus 2 is 4, there are infinitely many prime numbers, etc.) and, for example, we believe that Goldbach’s conjuncture makes sense, must be either true or false without there being any room

Classical Platonism vs. mathematical platonism

$$e^{i\pi} + 1 = 0$$

Platonism (*capital 'P'*)

- Eternal, Intelligible Forms;
- “Learning” as *anamnesis*
- Physical as shadow of the Real

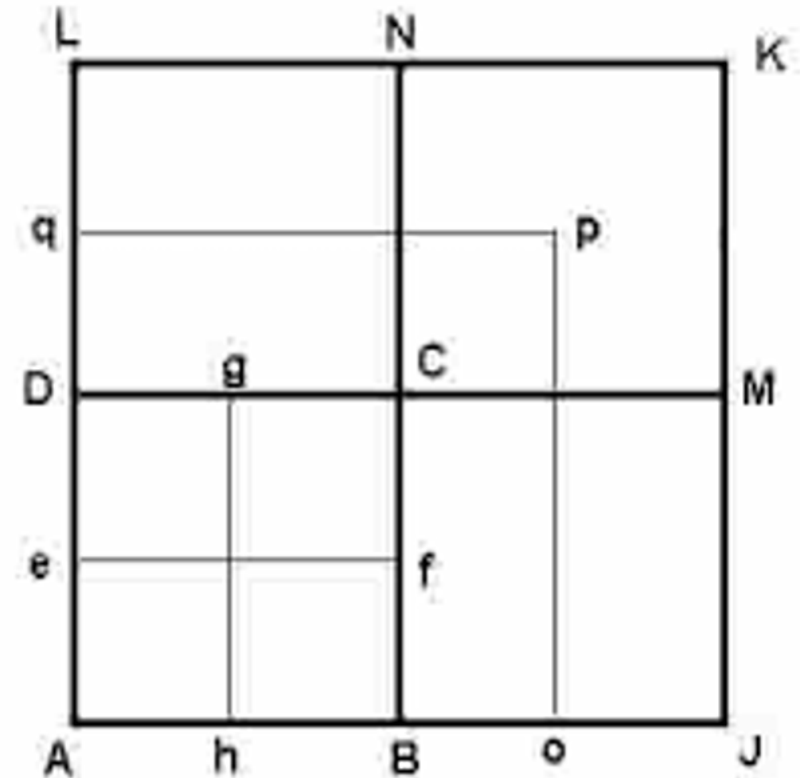


platonism (*lower case 'p'*)

- Objective mathematical truths
- “Aha!” or moment of proof
- “Unreasonable effectiveness”

What makes Plato's *Meno* a classic?

- ❖ Hilbert: "The art of doing mathematics is finding that special case that contains all the germs of generality."
-
- ❖ Meta-Meno Paradox: Learning that learning is Impossible?
 - ❖ Why the detour through rational estimates?
 - ❖ Chomsky "Plato's Problem": Poverty of Stimulus, Teaching often occurs spontaneously.
 - ❖ Karl Sigmund in *The Waltz of Reason* [2023]: Socrates initially talks, not about a square but about a rhombus.
 - ❖ Eugene Wigner "The Unreasonable Effectiveness of Mathematics in the Natural Sciences" [1960]





G. H. Hardy (1877-1947)

I remember once going to see him when he was ill at Putney. I had ridden in taxi, cab number 1729 and remarked that the number seemed to me rather a dull one, and that I hoped it was not an unfavorable omen.

1729

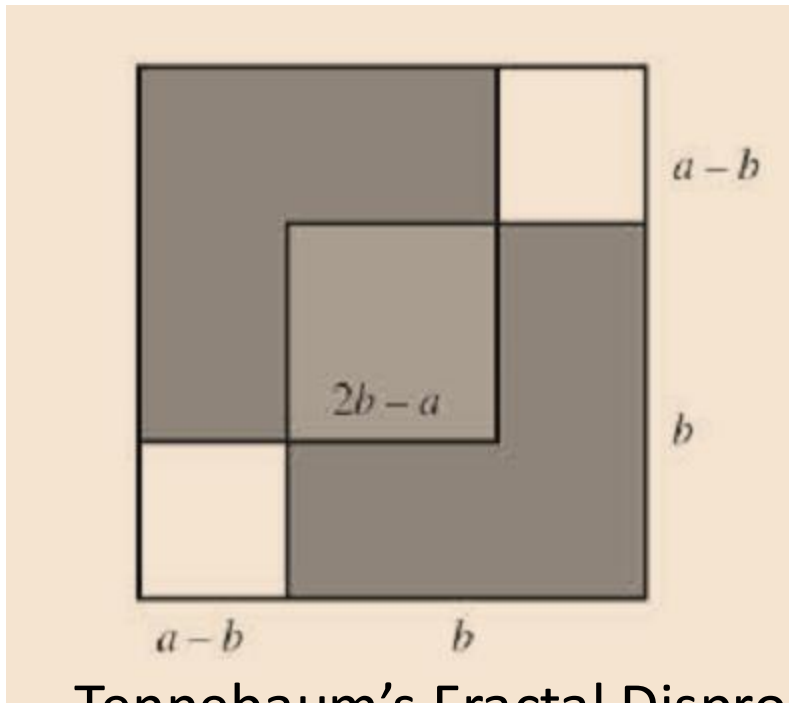
*Epistemology:
Romantic Mysticism
or Mathematical
Realism?*



S. Ramanujan (1887-1920)

"No," he replied, "it is a very interesting number; it is the smallest number expressible as the sum of two [positive] cubes in two different ways."

Foreshadowing fractals in the Foundations



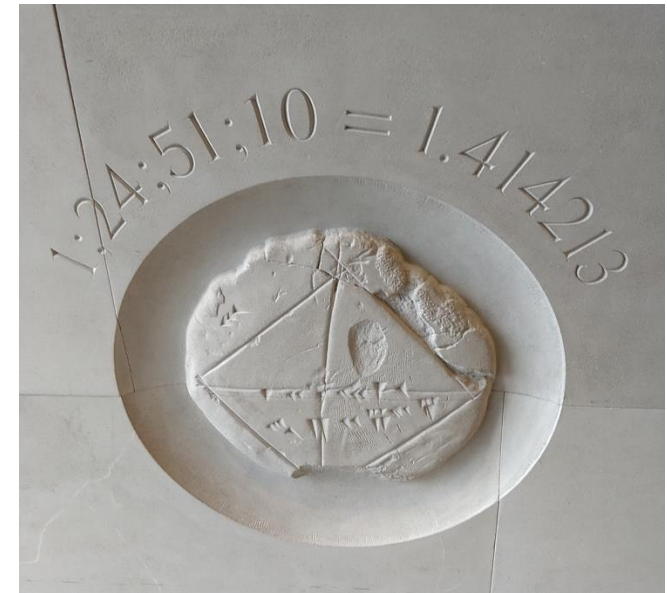
Tennebaum's Fractal Disproof

Fermat's Disproof by Infinite Descent

$\forall x[N(x) \wedge \forall y[N(y) \wedge y < x \rightarrow \neg F(y)] \rightarrow \neg F(x)] \rightarrow \forall x[N(x) \rightarrow \neg F(x)]$
 $\forall x[N(x) \wedge F(x) \rightarrow \exists y[N(y) \wedge y < x \wedge \neg F(y)] \rightarrow \forall x[N(x) \rightarrow \neg F(x)]$

x	$2/x$	average
1	2	1.5
1.5	1.33333333	1.41666667
1.41666667	1.41176471	1.41421569
1.41421569	1.41421144	1.41421356

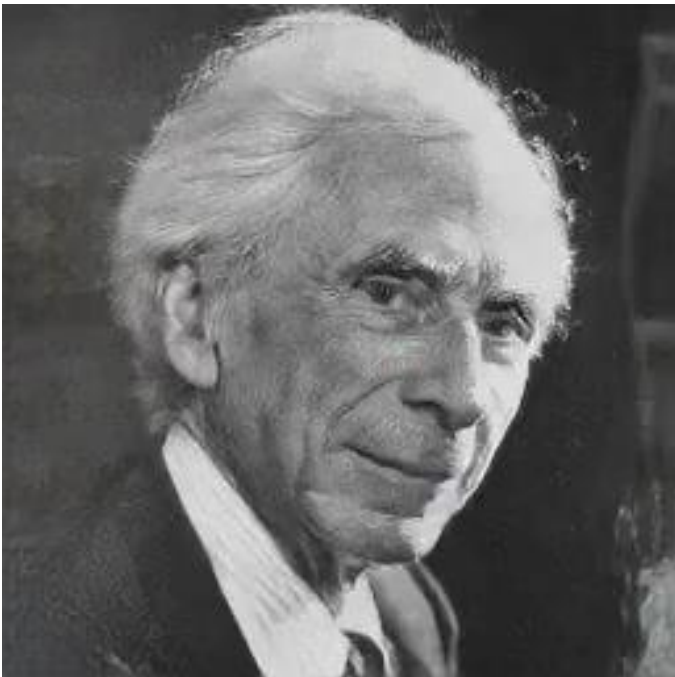
$$x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$$



The Babylonian clay tablet YBC 7289 (c. 1800–1600 B.C.) gives an approximation of the square root of 2 in four sexagesimal figures, 1 24 51 10 = 1.414213, which is accurate to six decimal digits. Photo by author of a reproduction from a mural in the Simon's Center for Geometry and Physics, Stony Brook University, Stony Brook, New York.

“Logic, being considered to be a branch of philosophy and having been treated by Aristotle, has been considered to be a subject only to be treated by those who are proficient in Greek. Mathematics, as a consequence, has only been treated by those who knew no logic. From the time of Aristotle and Euclid to the present century, this divorce has been disastrous.”

Four Paradoxes



LIAR PARADOX

Epimenides the Cretan said, “All Cretans are liars”.
Did Epimenides speak truly?

CANTOR’S POWERSSET THEOREM

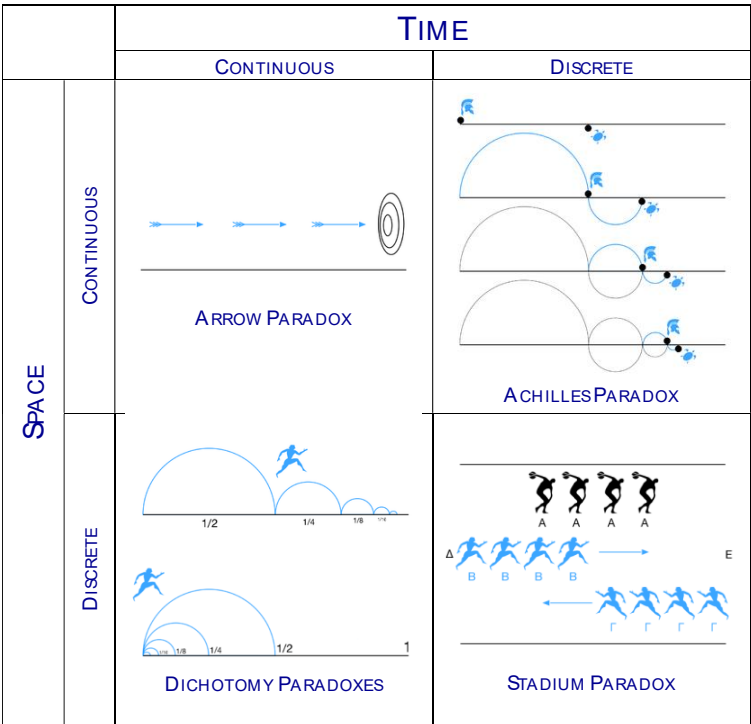
Cantor asked, “Can any set be the universe V of all sets?”
Assume there’s a 1-1 onto map $f: V \rightarrow \mathcal{P}(V)$.
Consider $D = \{x \in V \mid x \notin f(x)\}$

RUSSELL’S PARADOX

Russell asked Frege, “Does every condition specify a set?”
Consider $R = \{x \mid x \notin x\}$

BERRY PARADOX: “Does the specification ‘the least natural number not specifiable in less than 100 characters’ specifiable in less than 100 characters?”

Zeno’s Paradoxes



Zenoing the *Meno*

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

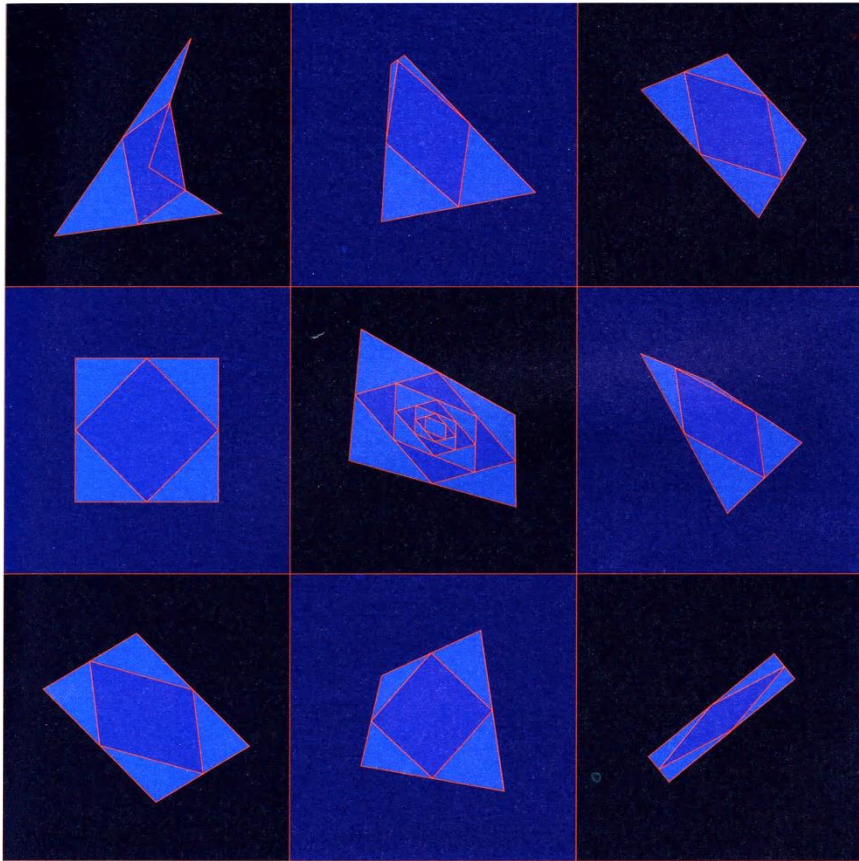
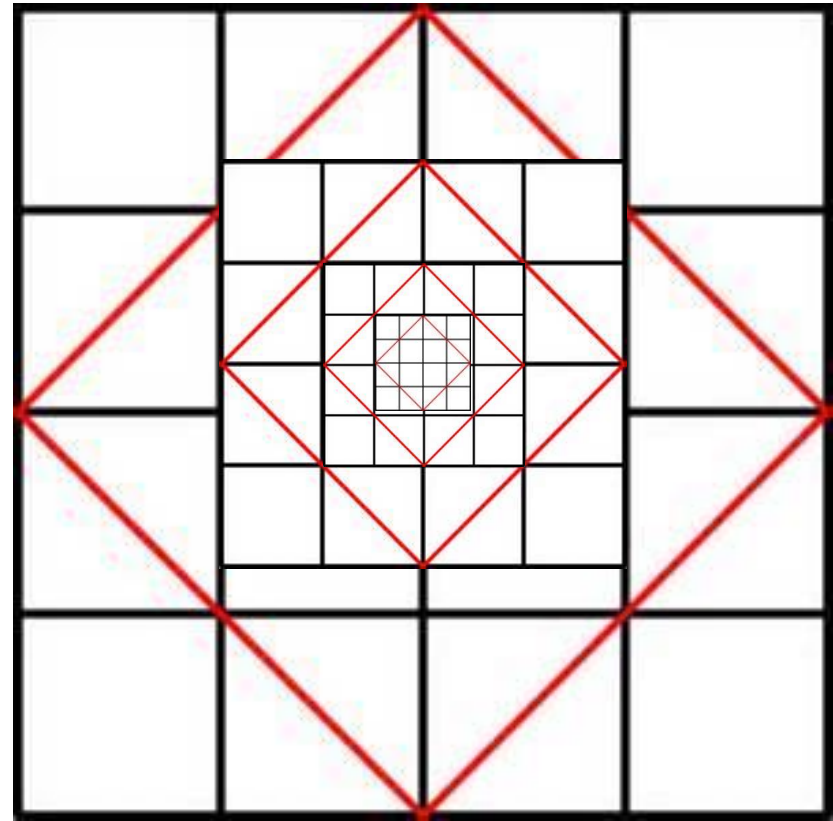
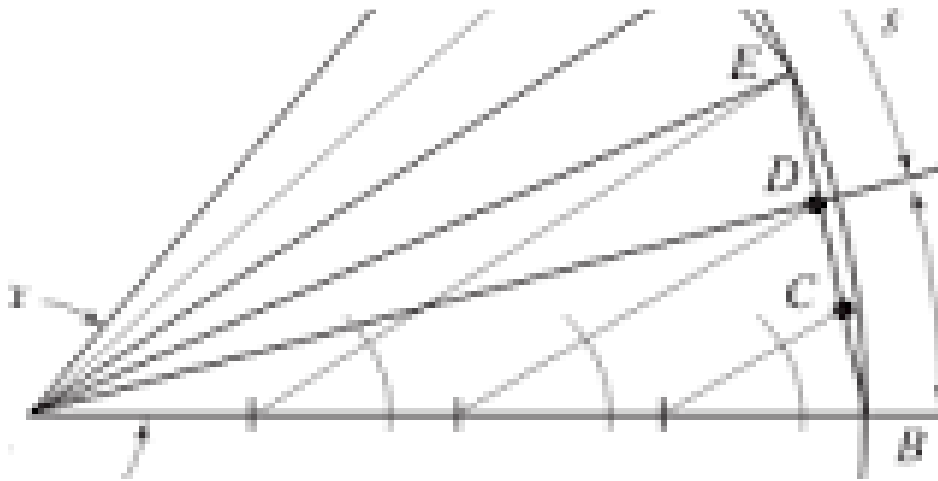
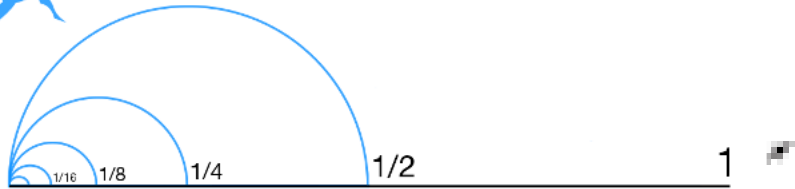
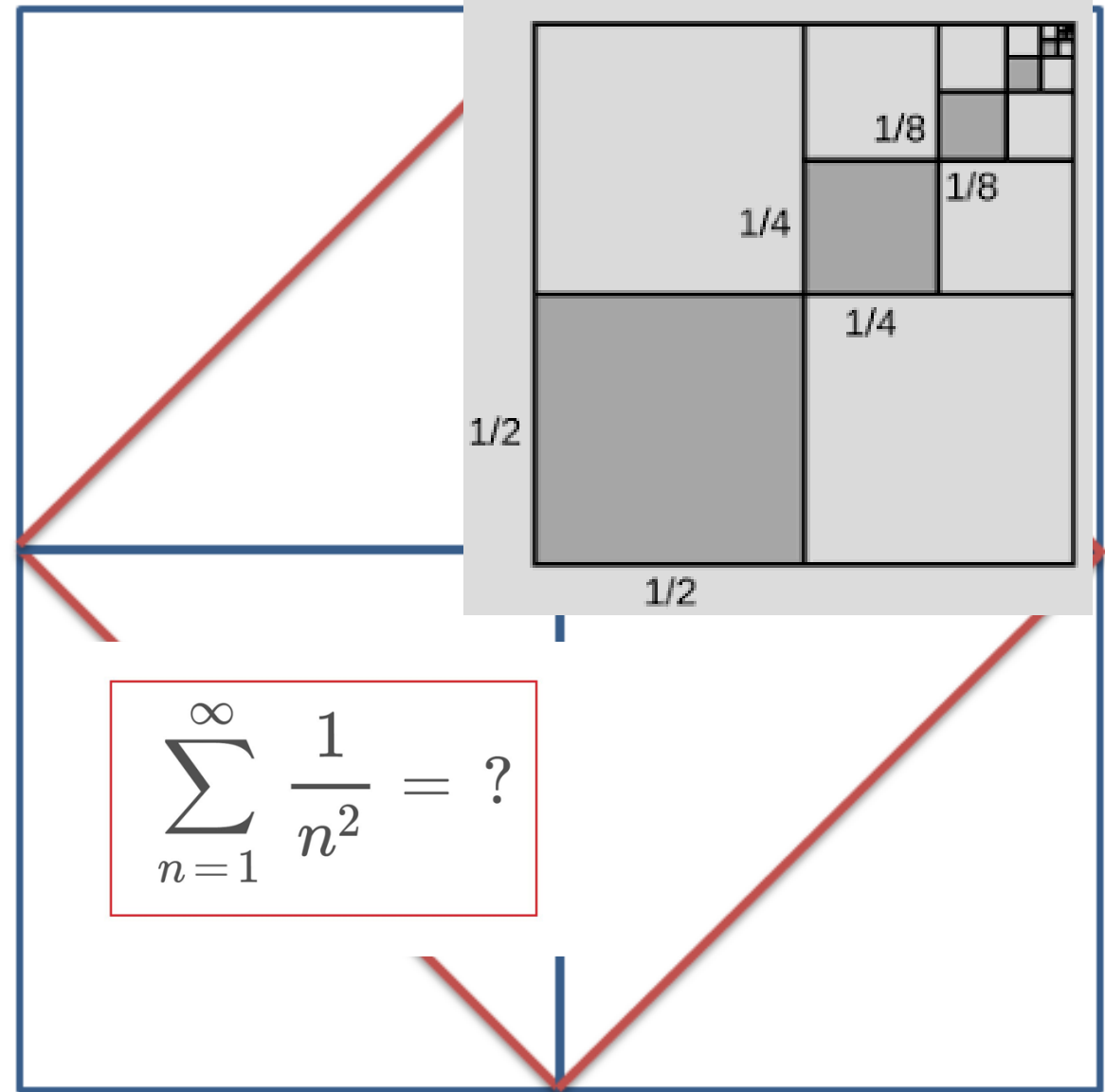


Plate 3. Quadrilaterals





A Zenoian Turing Machine (e.g., mentioned by Kripke and discussed by Shagrir) could complete infinitely many steps in a finite amount of time ,,,



Ian Stewart, "A Partly True Story" (signed Oct. 2023)

Scientific American, 268 (Feb. 1993), 110-112.



MATHEMATICAL RECREATIONS

by Ian Stewart

A Partly True Story

Allow me to introduce myself: Epimenides, professional liar. Well, that's not quite true. My name is really Herman Fenderbender, and I work for a car insurance company. But my friends at the Paradox Club call me Epimenides, and when I'm with them, I always lie.

Last Thursday it was raining, so I got to the club a bit late. Socrates and Plato were leaning against the bar, and next to them was a chubby little fellow.

"This is our newest member, Lukasiewicz," Plato chimed.

"Horried to meet you," I said in disgust. "My name's Zeno."

"He means he's delighted to meet you, and his name is Epimenides," Socrates explained. "Epimenides always tells lies."

"That's not true," I said. I opened my

wallet and took out my business card. "This isn't my card," I commented and handed it over. Lukasiewicz read one side of the card: *The sentence on the other side of this card is true.* He turned the card over and saw: *The sentence on the other side of this card is false.*

"Socrates is right, however. I always tell lies," I boasted.

Lukasiewicz shook my hand warmly. "It's one third false that I'm pleased to meet you, and both sides of your card are half true."

"Pardon?" I said.

"Lukasiewicz is interested in fuzzy logic," Plato explained.

"Instead of just the truth values 1 for a true statement and 0 for a false statement," Lukasiewicz said, "I am prepared to consider half-truths with truth value 0.5 or near-falsehoods with value

0.1—in general, any number between 0 and 1."

"Why would anyone want to do that?" I asked, bemused.

Lukasiewicz smiled. "Suppose I said the club president looks like Charlie Chaplin. Do you think that's true?"

"Of course not!"

"Not even his feet?"

"Well, I guess they do rather—"

"So it's not completely false, either."

"Well, he does look a bit like Chaplin."

Lukasiewicz leaned toward me. He had very penetrating eyes. "How much like him?"

"Around 15 percent I'd say."

"Good. Then my statement 'the club president looks like Charlie Chaplin' is 15 percent true. It has a truth value of 0.15 in fuzzy logic."

"That's just playing with words. It doesn't mean anything."

Lukasiewicz grasped my arm. "Oh, but it does. It helps to resolve paradoxes. For instance, you claim to be a complete liar. Let's think about your statement 'I am lying.' Or, more simply, 'this statement is false.' In classical logic, it is a paradox, yes? If it is true, then it is false; if it is false, then it is true. To put it another way, you have a statement P with truth value p , which is 0 or 1, and P says the truth value of this sentence is $1 - p$."

"Sorry, I didn't quite get that."

"Ah. If P is true, then its negation, not-P, is false, and its truth value is 0. And conversely. Now, $1 - 0 = 1$ and $1 - 1 = 0$, so if the truth value of P is p , then the truth value of not-P is $1 - p$."

"Precisely," Lukasiewicz said. "But it goes further than that. What we've been discussing is the beginnings of a whole new theory of dynamic logic, invented by Gary Mar and Patrick Grim in the department of philosophy of the State University of New York at Stony Brook. It provides a link between semantic paradoxes and chaos theory."

It was Socrates' turn to look puzzled.

"Oh, wake up. You know what chaos is. Simple deterministic dynamics leading to irregular, random-looking behavior. Butterfly effect. That stuff."

"Of course, I know that," Socrates said in irritation. "No, it was the idea of dynamic logic that was puzzling me. How can logic be dynamic?"

Lukasiewicz looked surprised. "How can it be anything else when discussing self-referential statements? The statement itself forces you to revise your estimate of its truth value. That revised value has to be revised again and again. Consider the Paradox of the Liar, your statement P: 'this statement is false.' Earlier I wrote an equation for its truth value: $p = 1 - p$. But what I should really

"Oh, I see."

"Right. Now the problem is that 'this statement' is P, so P is telling us that the truth value of P is $1 - p$. That's where the paradox comes from. If $p = 0$, then P tells us that $p = 1 - 0 = 1$. And if $p = 1$, then P tells us that $p = 1 - 1 = 0$. Neither choice is consistent."

I gave him a condescending smile. "Luke, all you've done is reformulate in complicated algebraic language what was obvious all along."

He smirked. "Maybe. But in fuzzy logic, there is a consistent solution to the equation $p = 1 - p$, namely, $p = 0.5$. So your claim to be a permanent liar is a half-truth, and everything works out fine. Your own statement leads inevitably to fuzzy logic." Plato slapped him on the back, and Socrates nearly knocked his cocktail over laughing. My face went red, but I saw the point.

"What about his business card?" Plato asked. Lukasiewicz was about to speak, but I stopped him. "Let me answer that. Seems to me I have two statements P and Q with truth values p and q . Moreover, P says Q is true, and Q says P is false. So the corresponding truth-value equations are

$$p = q \text{ for } P$$

$$q = 1 - p \text{ for } Q.$$

Those make no sense if p and q can only be 0 or 1. But there's a unique solution in fuzzy logic: $p = q = 0.5$. So each side of my card is a half-truth, and there's no paradox anymore."

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have written was a process that forces constant revision of your assessment of its truth value, $p \leftarrow 1 - p$. If you assume that P has a particular truth-value p , then P itself tells you to replace that truth value by $1 - p$. For example, if you started out thinking that P was 30 percent true, so that $p = 0.3$, the revision rule implies that $p = 0.7$, which in turn implies that $p = 0.3$ again...and you get an infinite sequence of truth values, oscillating periodically between the two values 0.3 and 0.7. The classical paradox, with $p = 0$ or 1, leads to the sequence 0, 1, 0, 1, ... which faithfully reflects the logical argument if P is false, then P is true, so P is false, so P is true, so ... The logical oscillations of the paradox are captured by the dynamics of the truth value."

"And $p = 0.5$ is the only value that doesn't lead to an oscillation," Plato mused.

"Precisely. Now, the Dualist Paradox on your business card is really a logical dynamic:

$$p \leftarrow q$$

$$q \leftarrow 1 - p.$$

Suppose you start out by estimating $p = 0.3$, $q = 0.8$. Then your first revision is to $p = 0.8$, $q = 0.7$. A further revision leads to $p = 0.7$, $q = 0.2$, a third to $p = 0.2$, $q = 0.3$. A fourth revision gives $p = 0.3$, $q = 0.8$, and you're back where you started. It cycles with period four—unless you start at $p = 0.5$, $q = 0.5$, when everything stays unchanged."

"Okay, I'll buy that," I said. "But what about the chaos?"

Lukasiewicz's face went very serious. "Before I can explain that, I must be more precise," he said. "If you want to play around with these ideas for yourself, I'd better tell you how to calculate fuzzy truth values for combinations of logical statements (see box at right). Although all you really need to know at the moment is that not-P has truth value $1 - p$ if P has truth value p . Second, you must know how to assess the truth value of statements about statements."

"I'd like an example," Socrates said.

"Okay. Suppose I said Plato is a good golfer. How true do you think that is?"

"Ooohhh—about 40 percent," Socrates said. Plato gave him a nasty look. "Well, Epimenides usually beats you, and he's pretty mediocre." I gave him an even nastier look.

"Fine. Let's call that statement S. It has a truth value $s = 0.4$. Suppose I make a statement about the statement S. Suppose I utter statement T: 'S is 100 percent true.' How true is statement T?"

Fuzzy Logic

In classical logic, a statement I has a truth value of either 1 for true or 0 for false. The statement "the sun is shining" has a truth value of 0 if it is cloudy, in general, statement P has a truth value p equal to 1 or 0. In fuzzy logic, a statement can have a truth value of between 1 and 0. If a cloud obscures a quarter of the sun, then statement P has a value of 0.25.

In fuzzy logic, like the classical theory, the truth value of a statement will change when applying the operators NOT, AND, OR, IMPLIES and IF AND ONLY IF.

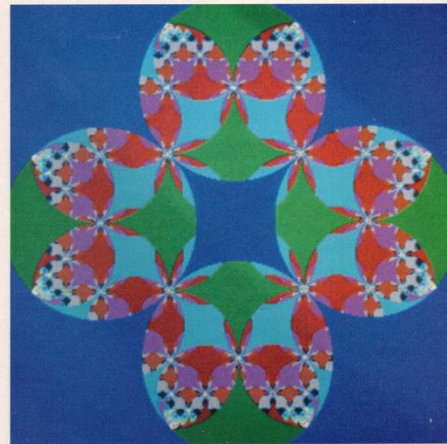
NOT-P has a truth value of $1 - p$. EXAMPLE: If the sun is shining with a truth value of 0.25, then the sun is NOT shining with a truth value of 0.75.

P AND Q has a truth value equal to the lesser of p and q where q is the truth value of statement Q. EXAMPLE: The sun is shining with a truth value of 0.25, AND Jane is getting tan with a truth value of 0.10. The value of the example is 0.10.

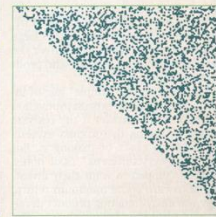
P OR Q has a truth value equal to the greater of p and q . EXAMPLE: The sun is shining with a truth value of 0.25, OR Jane is getting tan with a truth value of 0.10. The value of the example is 0.25.

P IMPLIES Q has a truth value equal to the lesser of 1 and $1 - p + q$. EXAMPLE: If the sun is shining with a truth value of 0.25, then Jane is getting tan with a truth value of 0.10. The value of the example is 0.85.

P IF AND ONLY IF Q has a truth value equal to $1 - |p - q|$, that is, one minus the absolute value of p minus q . EXAMPLE: The sun is shining with a truth value of 0.25 IF AND ONLY IF Jane is getting tan with a truth value of 0.10. The value of the example is 0.85.



DIAGRAMS OF "ESCAPE TIME" (left) and a logical attractor (right) were created to analyze a self-referential statement. Such statements typically lead to paradoxes in classical logic.

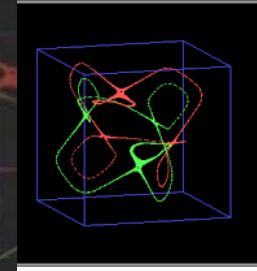


The illustrations above are based on the sentence "the assessed falsehood of this statement is not different from its assessed truth."

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The Philosophical Computer

Exploratory Essays in Philosophical Computer Modeling



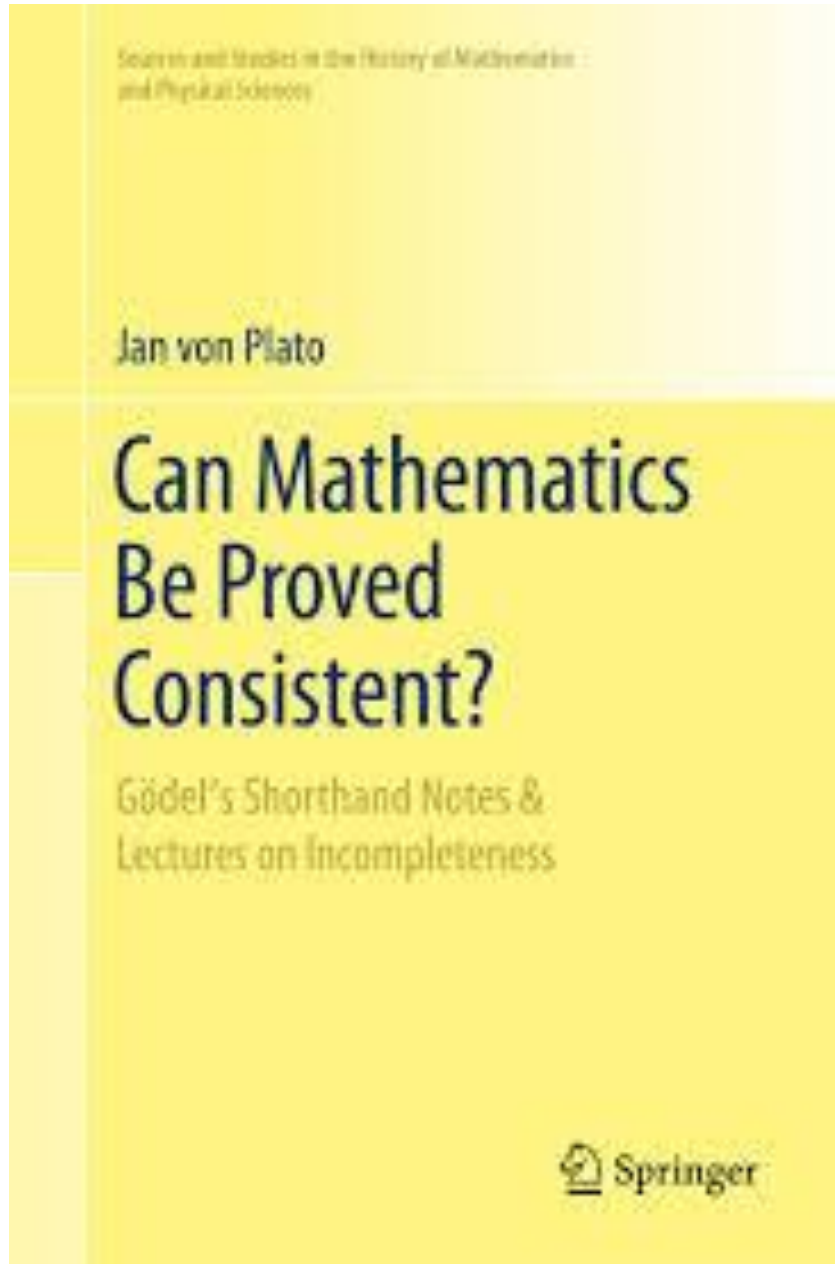
Patrick Grim, Gary Mar, and Paul St. Denis
with the Group for Logic and Formal Semantics

Includes CD-ROM for Macintosh and Windows
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“a landmark ... visible far in space and time”

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- ❖ AWAKENING FROM A DREAM?





September 7, 1930 on the last day of the *second Tagung für Erkenntnislehre der exakten Wissenschaften* in Königsberg, Gödel:
“One can, assuming the consistency of classical mathematics, even give examples of propositions (and in fact of the kind of Goldbach and Fermat) that really are contentually true but are unprovable in the formal system of mathematics.”

September 8, 1930 Hilbert’s address in Königsberg to the Society of German Natural Scientists and Physicians:
Wir müssen wissen, wir werden wissen!

November 20, 1930 John von Neumann’s letter to Gödel:
“Using the methods you employed so successfully . . . I achieved a result that seems to me to be remarkable, namely, I was able to show that the consistency of mathematics is unprovable.”

Provability Logic

Gödel [1933o]

(T) $\Box\varphi \rightarrow \varphi$

(K) $\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$

(4) $\Box\varphi \rightarrow \Box\Box\varphi$

Henkin [1952]:

Is the *positive* Gödel sentence provable?

Martin Löb [1955]00

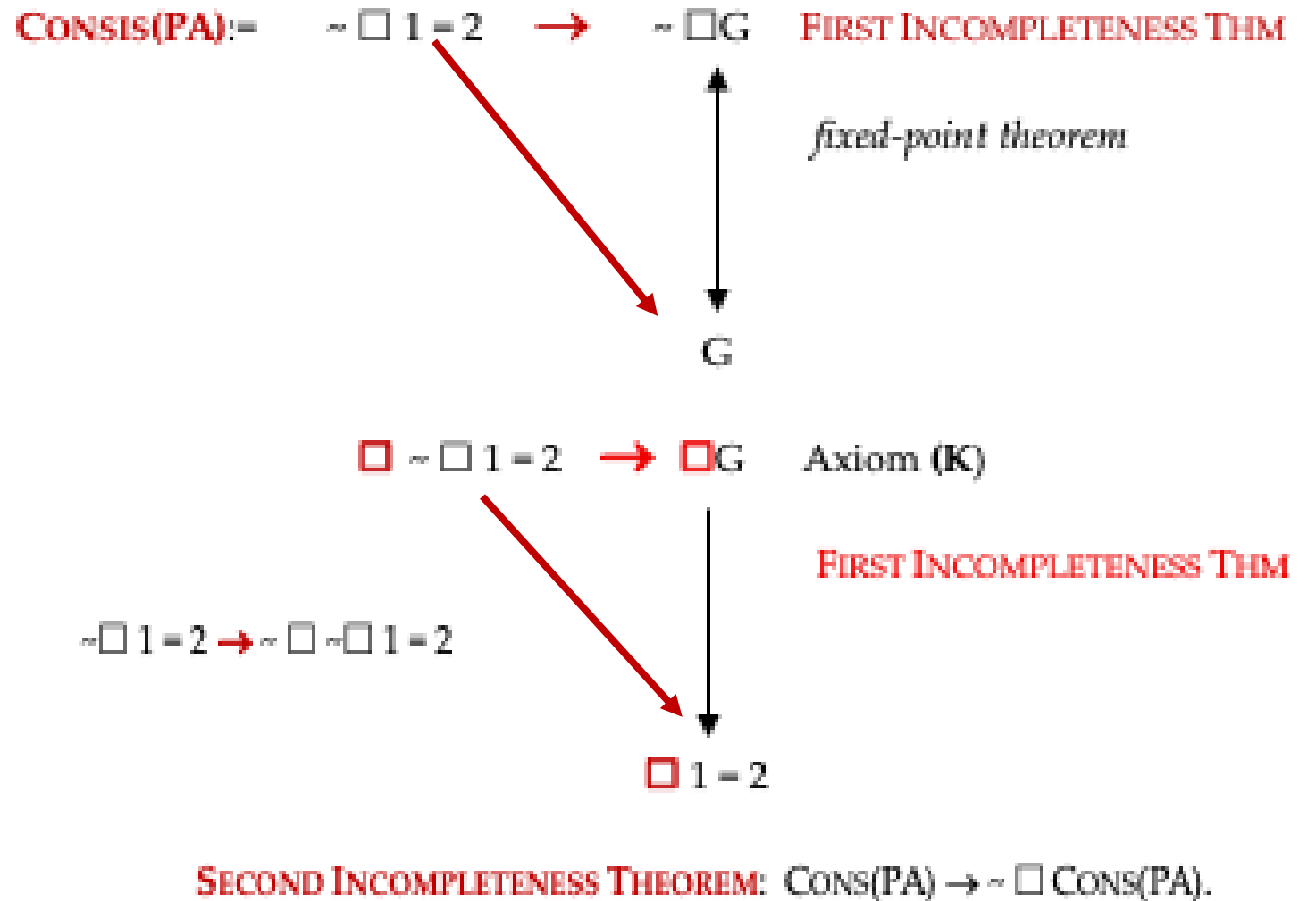
(L) $\Box(\Box\varphi \rightarrow \varphi) \rightarrow \Box\varphi$

- | | | |
|-----|---|--|
| 1. | $\sigma \leftrightarrow (\Box\sigma \rightarrow \varphi)$ | fixed-point theorem |
| 2. | $\Box\sigma \rightarrow \Box(\Box\sigma \rightarrow \varphi)$ | necessitation, (K) |
| 3. | $\Box\sigma \rightarrow (\Box\Box\sigma \rightarrow \Box\varphi)$ | necessitation, (K) |
| 4. | $\Box\sigma \rightarrow \Box\Box\sigma$ | (4) |
| 5. | $\Box\sigma \rightarrow \Box\varphi$ | CD, 3, 4 MP |
| 6. | $\Box\sigma \rightarrow \varphi$ | using $\vdash \Box\varphi \rightarrow \varphi$ |
| 7. | $(\Box\sigma \rightarrow \varphi) \rightarrow \sigma$ | 1, BC |
| 8. | σ | 6, 7 MP |
| 9. | $\Box\sigma$ | 8, necessitation |
| 10. | φ | 6, 9 MP |

2nd Incompleteness Theorem

The hope of finding “...a proof for freedom from contradiction undertaken by Hilbert and his disciples” has “vanished entirely in view of some recently discovered facts. It can be shown quite generally that there can exist no proof of the freedom of contradiction of a formal system S which could be expressed in terms of the formal system S itself”

(“The present situation in the foundations of mathematics” [*1933o], GCW-III, 52). Cambridge Massachusetts, Invited lecture to joint meeting of the Mathematical Association of America and the American Mathematical Society.



“ ... the human mind (even within the realm of pure mathematics) infinitely surpassed the power of any finite machine, or else there exists absolutely unsolvable Diophantine problems.... ”

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Intellectual Honesty

Gödel's 1933 lecture "The present situation in the foundations of mathematics", December 1933: *"The result of the preceding discussion is that our axioms, if interpreted as meaningful statements, necessarily presuppose a kind of Platonism, which cannot satisfy any critical mind and which does not even produce the conviction that they are consistent."* [*1933o, GCW III, 50]

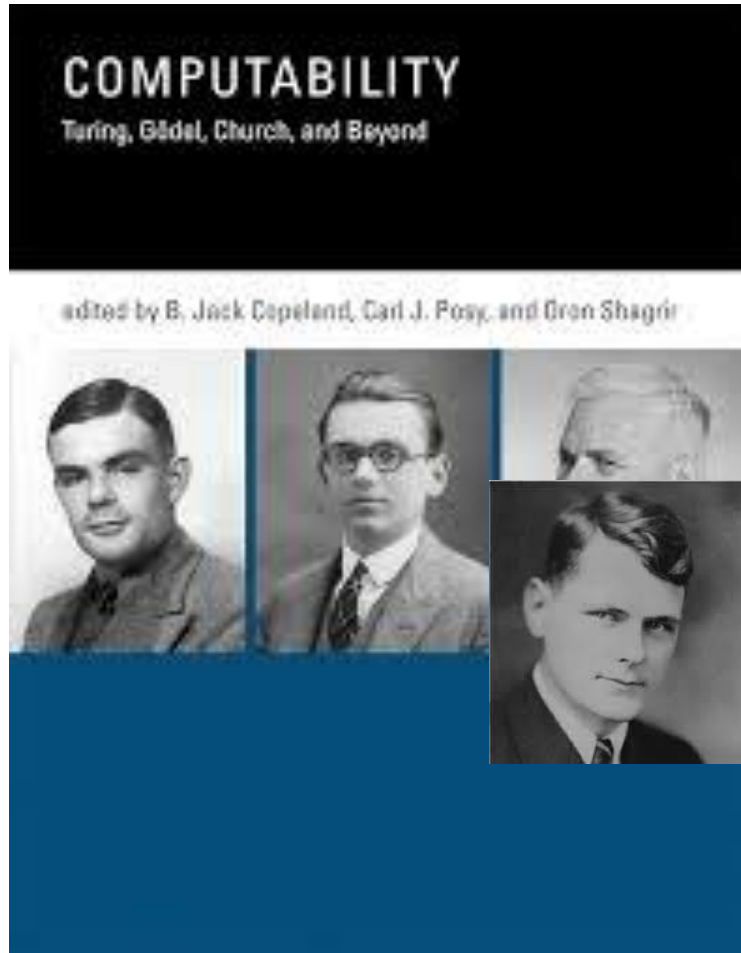
Roger Penrose ([1994], 21): "It seems likely that he [Turing] viewed physical action in general—which would include the action of a human brain – to be always reducible to some kind of Turing-machine action."

Turing in his BBC radio broadcast "Can Digital Computers Think?" (15 May 1951) calls attention to Eddington's view that the "indeterminacy principle in quantum mechanics" might lead to actions not Turing-machine computable." (See Copeland, "Physics and uncomputability: a brief history"

(B&vL [2012], 652ff.)



Church-Turing Thesis



Gödel, Princeton lecture [1946] that “the great importance of the concept of general recursiveness (or Turing’s computability) is largely due to the fact that **with this concept one has for the first time succeeded in giving an absolute definition of an interesting epistemological notion,** i.e., one not depending on the formalism chosen. . . . For the concept of computability, however, although it is merely a special kind of demonstrability or decidability, the situation is different. **By a kind of miracle it is not necessary to distinguish orders, and the diagonal procedure does not lead outside the defined notion.**” [Gödel, CW II, p.150]

Turing[1936/7] TMs are Abstract Objects

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO
THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

The "computable" numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means. Although the subject of this paper is ostensibly the computable numbers, it is almost equally easy to define and investigate computable functions of an integral variable or a real or computable variable, computable predicates, and so forth. The fundamental problems involved are, however, the same in each case, and I have chosen the computable numbers for explicit treatment as involving the least cumbersome technique. I hope shortly to give an account of the relations of the computable numbers, functions, and one to another. This will include a development of the theory of functions of a real variable expressed in terms of computable numbers. According to my definition, a number is computable if its decimal can be written down by a machine.

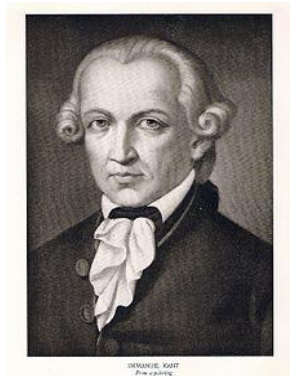
Turing's Halting Problem:

1 if $P(\#P)$ does not halt

$H(\#P) = \{$

0 if $P(\#P)$ does halt

	$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	...
M_1	1	reject	accept	reject	0
M_2	accept	0	accept	1	accept
M_3	reject	reject	0	reject	1
M_4	accept	accept	reject	0	reject
...	0	1	1	1	0



“The light dove, in free flight cutting through the air the resistance of which it feels, could get the idea that it could do even better in airless space. Likewise, Plato abandoned the world of the senses because it posed so many hindrances for the understanding and dared to go beyond it on the wings of the ideas, in the empty space of pure understanding.”

— IMMANUEL KANT, *Critique of Pure Reason* [1781, 1787]



*Platonism without computationalism is epistemologically blind;
Computationalism without platonism is theoretically empty.*



- ❖ Gödel’s proofs of the *a priori*—from Arithmetic’s Incompleteness Theorems to the need to search for new of infinity for ZF set theory—is a defense of the *autonomy* of philosophy.
- ❖ The “unreasonable effectiveness” of Turing’s Machines in modelling *a posteriori* processes—from Algorithms playing chess to painting the Zebra’s stripes—proves the *applicability* of philosophy.



Dualism is picturable but implausible: "picture thinking" places two domains side-by-side, but their interaction is mediated by a mind ...



Murmuration of Starlings

(click on image)



Boids Algorithm

(click on image)

“Time has no specific character of being.”

MATHEMATICAL REALISM

MINDS VS. MACHINES?

MODEL PROVABILITY LOGIC

THE UNREALITY OF TIME

GOD'S GOD



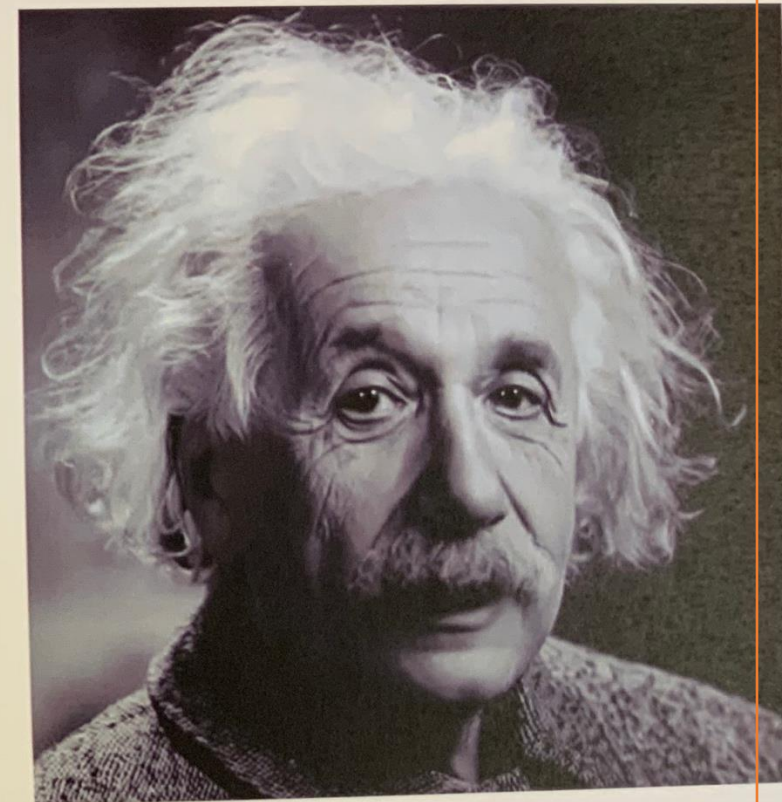
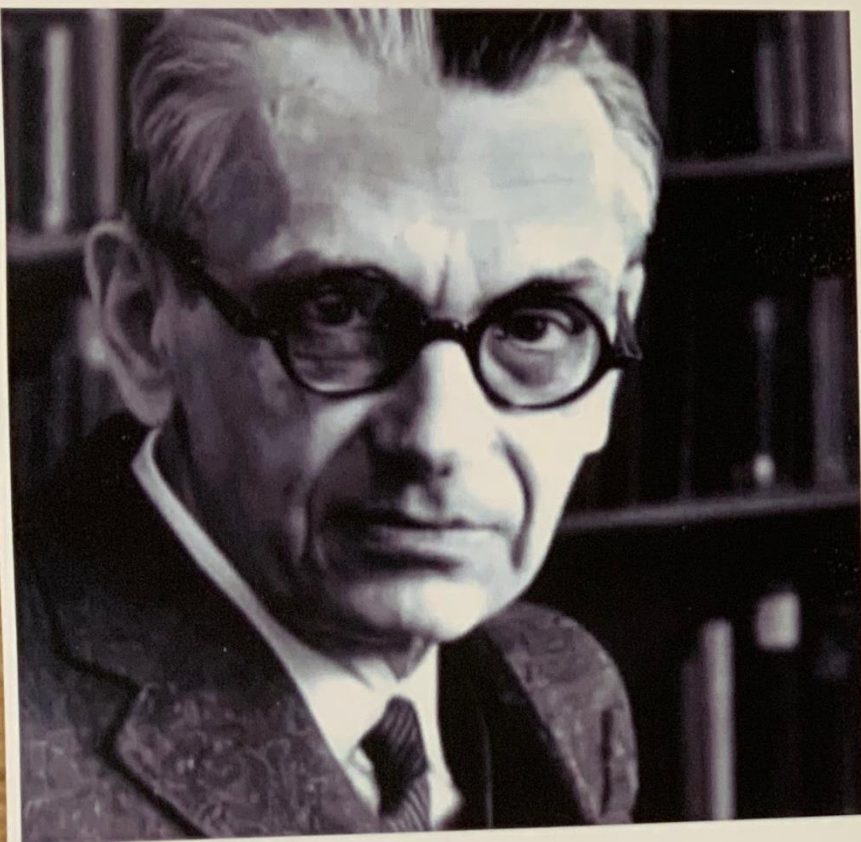


"Time has no specific character of being. In relativity theory the temporal relation is like far and near in space. I do not believe in the objectivity of time. The concept of *Now* never occurs in science itself, and science is supposed to be concerned with all that is objective."



The introductory paragraph for his essay for Einstein [1949a]:

"One of the most interesting aspects of relativity theory for the philosophical-minded consists in the fact that it gave new and surprising insights into the nature of time, of that mysterious and seemingly self-contradictory¹ being which, on the other hand, seems to form the basis of the world and our existence.



Time is objective $\Leftrightarrow$ Ordering of moments of Time is Total:

Irreflexive: $\forall x \sim x < x$

Transitive $\forall xyz (x < y < z \rightarrow x < z)$

Connected $\forall xy (x \neq y \rightarrow x < y \vee y < x)$

What modal axiom connects modality with actuality?

What modal axiom connects modality with actuality?

No. 4016 October 19, 1946

NATURE

LETTERS TO THE EDITORS

The Editors do not hold themselves responsible for opinions expressed by their correspondents. No notice is taken of anonymous communications.

Rotating Universe?

ONE of the most mysterious results of the astronomical studies of the universe lies in the fact that all successive degrees of accumulation of matter, such as planets, stars and galaxies, are found in the state of more or less rapid axial rotation. In various cosmological theories the rotation of planets has been explained as resulting from the rotation of stars from which they were formed. The rotation of stars themselves (in particular that of B-stars) can be presumably reduced to their origin from the rotating gas-masses which form the spiral arms of various galaxies. But what is the origin of galactic rotation?

If, according to the current theories, we consider the galaxies as the result of gravitational instability of the originally uniform distribution of matter in space, we will find it very difficult to understand why such condensations are in most cases found in the state of rather fast rotation. In fact, on the basis of statistical distribution of angular momentum, we would rather expect such condensations to show no more rotation than the water droplets in a fog formed from over-saturated vapour. Barring the possible explanation of the rotation of galaxies on the basis of the alleged irregular turbulent motion of the masses of the universe, we can ask ourselves whether it is not possible to assume that all matter in the visible universe is in a state of general rotation around some centre located far beyond the reach of our telescopes?

The answer to such, at first sight fantastic, question need not wait until much larger telescopes shall have been built. It can be, in fact, settled by present means of observation. We know that the rotation of the stars of our system around the galactic centre can be proved by the study of the so-called Doppler-effect in the radial velocities of comparatively near stars. In fact, due to the phenomenon of differential rotation, the mean radial velocities of stars located along the galactic plane show a double-sine periodicity with nodal axes directed parallel and perpendicular to the line connecting the sun with the centre of rotation. Thus if the realm of galaxies as seen through Hubble's telescope represents only a small part of a much larger system (a 'super-galaxy' in the super-Shapley sense) rotating around a distant centre, careful observations of mean radial velocities of galaxies located in different regions of the sky should reveal similar periodicity. The existence of this effect would prove general rotation of the universe and indicate the direction towards the rotation centre without, however, giving us its distance. Thus, it seems that the answer to the problem of universal rotation lies within the grasp of modern astronomical techniques.

It must be added in conclusion that in the language of the general theory of relativity such a rotating universe can be probably represented by the group of anisotropic solutions of the fundamental equations of cosmology.

Department of Physics,
George Washington University,
Washington, D.C.
Sept. 13.

G. GAMBOL

total field H , $f_h = eH/4\pi cm. = 1.92 H$ Mc. These can lower limit to the frequency of the radiations which can the earth, and their validity has been verified by measurements.

If we apply these conditions to the sun, and also to 1 and at once that severe physical conditions have to be the emission of radio-waves from these bodies. Taking this we should have

$$\begin{aligned} N &< 1.25 \times 10^{-4} \cdot f^2 \\ &< 1.25 \times 10^4 \text{ for } f = 10 \text{ Mc.} \\ &< 5 \times 10^4 \text{ for } f = 300 \text{ Mc.} \end{aligned}$$

The concentration of electrons in the different layers has been found by well-tried astrophysical methods* to be values of 10^{18} per c.c. for the reversing layer, 4×10^{18} per mean chromosphere, and 4×10^{19} per c.c. for the base corona. It is, therefore, obvious that α -radiations of mid range which we obtain from the sun cannot have their in the reversing layer or the chromosphere, but only in and that also progressively in the outer layers as the is increased. But the corona has been shown to be a 'gas atmosphere' without any heavier atomic particles, or small concentrations of heavily ionized Fe, Si and Ca in the coronal lines. The mechanism of origin contemplate Stein, Henney and Keenan¹ which ascribes the radio-wave emission between protons and electrons therefore appears the ground in the case of the sun.

The α -waves. For the α -waves, the value of f_h in decies varies from 46 Mc. for the quiescent sun to roughly 4,000 spot, taking $H = 3,000$. These are frequencies of an order not contemplated in Appleton's theory, but a little word whatever has been said regarding the α -wave also applies to which corresponds to the condition $f_e(f_e - f_h) > 8 \cdot 10^4 \times N$ emphasis. In fact, this wave cannot escape unless f_e is value, > 46 Mc. The α -wave corresponds to $f_e(f_e - f_h) > 8 \times 10^4 N$.

The possibility of reception of this wave on the earth has been noticed by European and American workers, but obtained distinctly on several occasions by Yoshida² and his findings have been confirmed by Lely Bursani³ and R. K. Bursani⁴ have shown that any radio-wave on the earth would be decomposed into three waves a Zeeman effect, the p-component corresponding to the the S-components to the α -waves. If this deduction be at once seen that for the spots, the α -wave of this type has probability of escape; for now we should have

$$\begin{aligned} N &< 1.25 \times 10^4 f_e(f_e + f_h) \\ &< 1.25 \times 10^4 f_e f_h, \text{ taking } f_h \gg f_e \\ &< 5 \times 10^4 \text{ for } 10 \text{ Mc. waves, and } < 10^{10} \text{ for } \alpha \text{ waves;} \end{aligned}$$

taking $f_h = 4,000$ Mc., corresponding to the field-strength of a quiescent sun, the figures are $N < 8 \times 10^4$ and respectively. Hence the probability of escape of these waves from a quiescent sun continues to be very small. If the wave, the deeper layers. For larger spots, the field generally has been known to reach values as high as 4,000 gauss. From these arguments, it is fair to draw the conclusion large spots are just the regions whence the α -waves of 10 range 10-200 Mc. can escape. The value of the field corresponds to the level where the atomic lines originate

McTaggart “The Unreality of Time” [1908]

McTaggart [1908] in “The Unreality of Time”:

“If one of the determinations past, present and future can ever be applied to [an event] then one of them has always been and always will be applicable, though of course not always the same one.”

$\blacklozenge P \rightarrow \blacksquare(\lozenge P \vee P \vee \blacklozenge P)$	Connectedness of the past
$\blacklozenge P \rightarrow \square \blacklozenge P$	Transitivity of “earlier than”
$\lozenge P \rightarrow \blacksquare \lozenge P$	Transitivity of “later than”
$\lozenge P \rightarrow \square(\lozenge P \vee P \vee \blacklozenge P)$	Connectedness of the future

McTaggart’s Argument:

- (M₁) If time is real, then time is dynamic and hence is an A-series.
- (M₂) However, there can be no A-series because the A-series conception is logically inconsistent.
 - (A₁) According to the A-series conception, events can have the A-series properties of being future, present, and past;
 - (A₂) but these A-series properties are mutually exclusive.
- (M₃) $\therefore$ Time is unreal.



J. M. E. McTaggart
(1866 – 1925)



G. E. Edward Moore
(1873 – 1958)



Bertrand Russell
(1872 – 1970)



*“My belief is theistic. not pantheistic,
following Leibniz rather than Spinoza....”*

“Visited Gödel at the Institute today. A true miracle: he has gained 18 pounds, is in best shape, sparkling conversation. Lots about politics but then about his ontological proof—he had the result several years ago, is now happy with it but hesitates over publishing it. It would be concluded that he really believes in God, whereas he is only undertaking a logical investigation (*i.e.*, he shows that such a proof, appropriately axiomatized, is possible under classical assumptions—perfections, *etc.*). I joked that he should use a pseudonym—but he has already told Hempel and Scott about it and I said people would recognize ‘the claw of the lion’, as they did with Newton.”

—Oskar Morgenstern’s diary 29 August 1970



GÖDEL AX. 1

"The conjunction of positive properties is positive."

GÖDEL AX. 2

Scott's Formulation [1970]

Anderson's Emendation [1990]

Anderson & Getting Emendation [1996]

GÖDEL DEF. 1

"A being is god-like iff it has all positive properties."

GÖDEL DEF. 2

SCOTT'S AMENDMENT

" ϕ is an essence of x iff x has ϕ and x has any property necessarily entailed by ϕ "

GÖDEL AX. 3

"A positive (non-positive) property is necessarily positive (non-positive)."

Gödel note: " $\Diamond \exists x Gx$ means the system of all positive properties is compatible...."

$P(G)$, i.e., god-likeness is positive AX. 5

$G(x) \rightarrow \Box \exists x Gx$ THM 2.
 $\therefore \Diamond Gx \rightarrow \Diamond \Box \exists x Gx$

There exists a (unique) god-like being.

$P(\phi) \wedge P(\psi) \rightarrow P(\phi \wedge \psi)$

$P(\phi) \vee P(\sim \phi)$

$P(\phi) \leftrightarrow \sim P(\sim \phi)$

$P(\phi) \rightarrow \sim P(\sim \phi)$

$\sim [P(\Box \phi) \leftrightarrow P(\sim \Box \phi)]$

$Gx \leftrightarrow \Box \forall \phi [P(\phi) \rightarrow \phi x]$

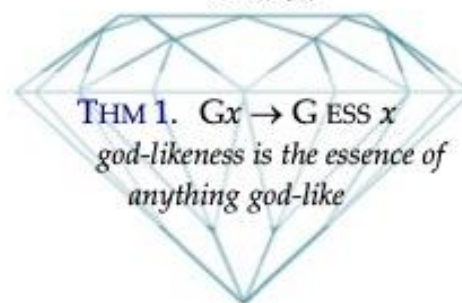
$\phi \text{ ESS } x \leftrightarrow \phi(x) \wedge [\Box \forall x (\phi x \rightarrow \psi x) \rightarrow \psi x]$

$P(\phi) \rightarrow \Box P(\phi)$

$\sim P(\phi) \rightarrow \Box \sim P(\phi)$

$P(\phi) \wedge \Box \forall x (\phi x \rightarrow \psi x) \rightarrow P\psi$

$\Diamond \exists x Gx$



THM 3. $\exists x Gx \wedge \Box \exists x Gx$

Gödel footnote: "And for any number of summands"

Gödel note: "exclusive 'or'"

"A property is positive if and only if its negation is not positive."

"The negation of a positive property is not positive."

"Exactly one of 'being necessarily ϕ ' and 'not being necessarily ϕ ' is positive."

Gödel note: "God"

Gödel notebooks: "For this is it required that all the properties of God are defined by a second-order property" GCW-III, 431

Gödel note: "Essence of x ";

Gödel footnote: "any two essences of x are necessarily equivalent"

Gödel: "because it follows from the nature of the property"

Gödel notebook: "That the necessity of a positive property is positive is the essential presupposition of the ontological proof."

(GCW-III, 435)

"Anything entailed by a positive property is positive."

$Ex \leftrightarrow \forall \phi [\phi \text{ ESS } x \rightarrow \Box \exists x \phi x]$

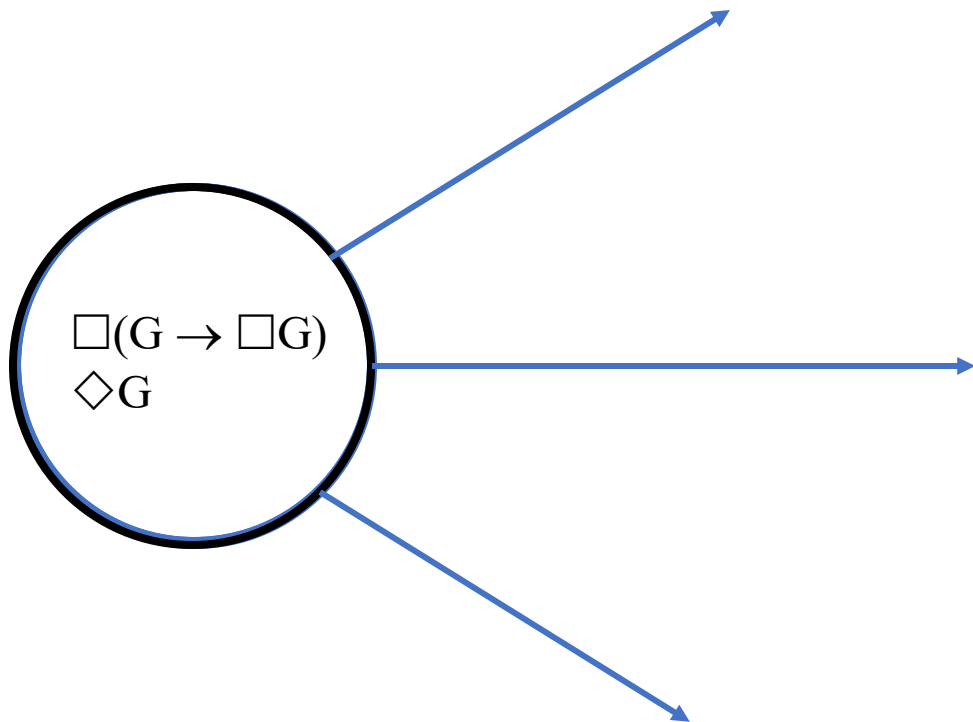
DEF. 3 Gödel: "necessary Existence)

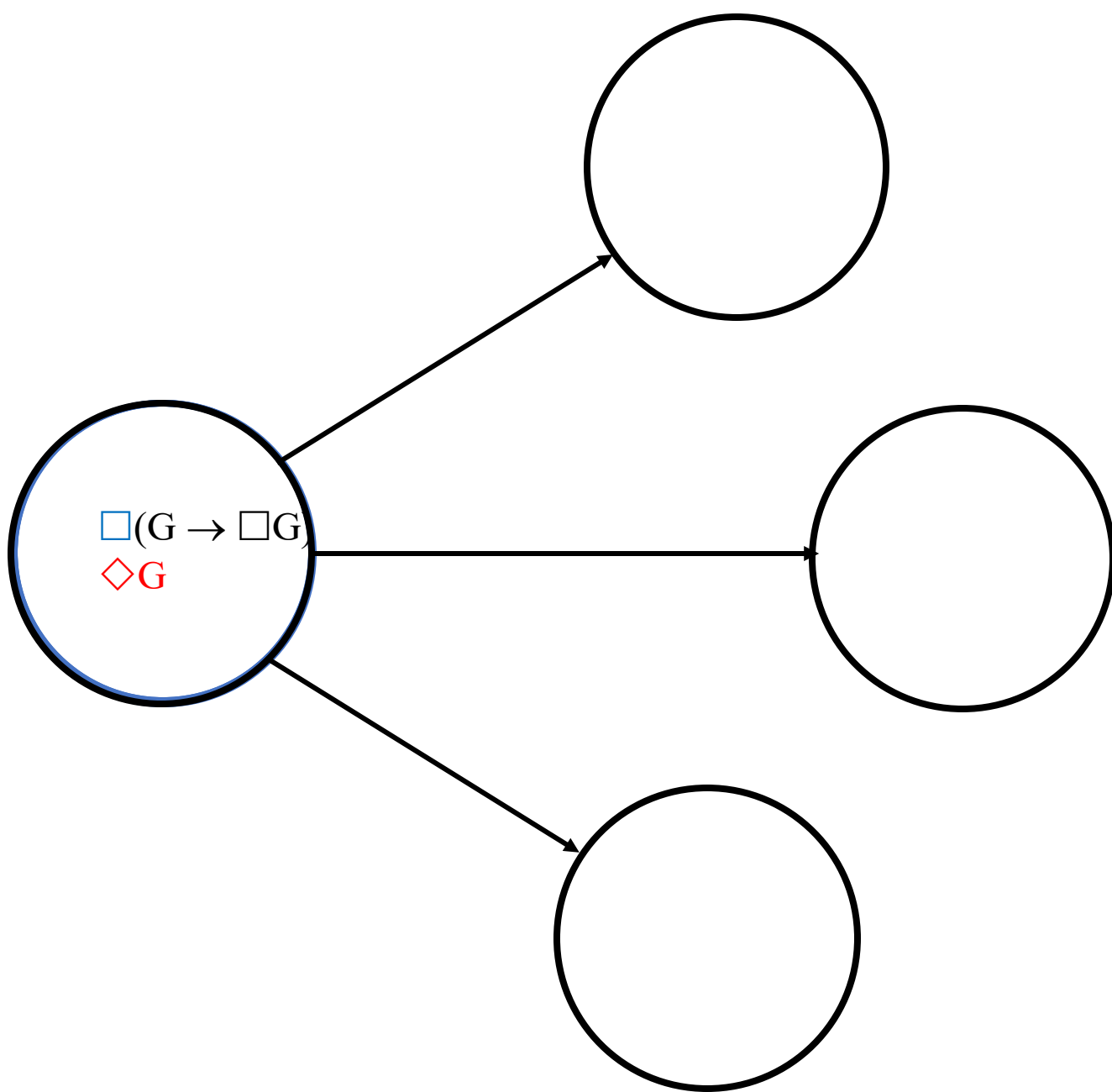
"Something has necessary existence iff every essence of x is necessarily instantiated."

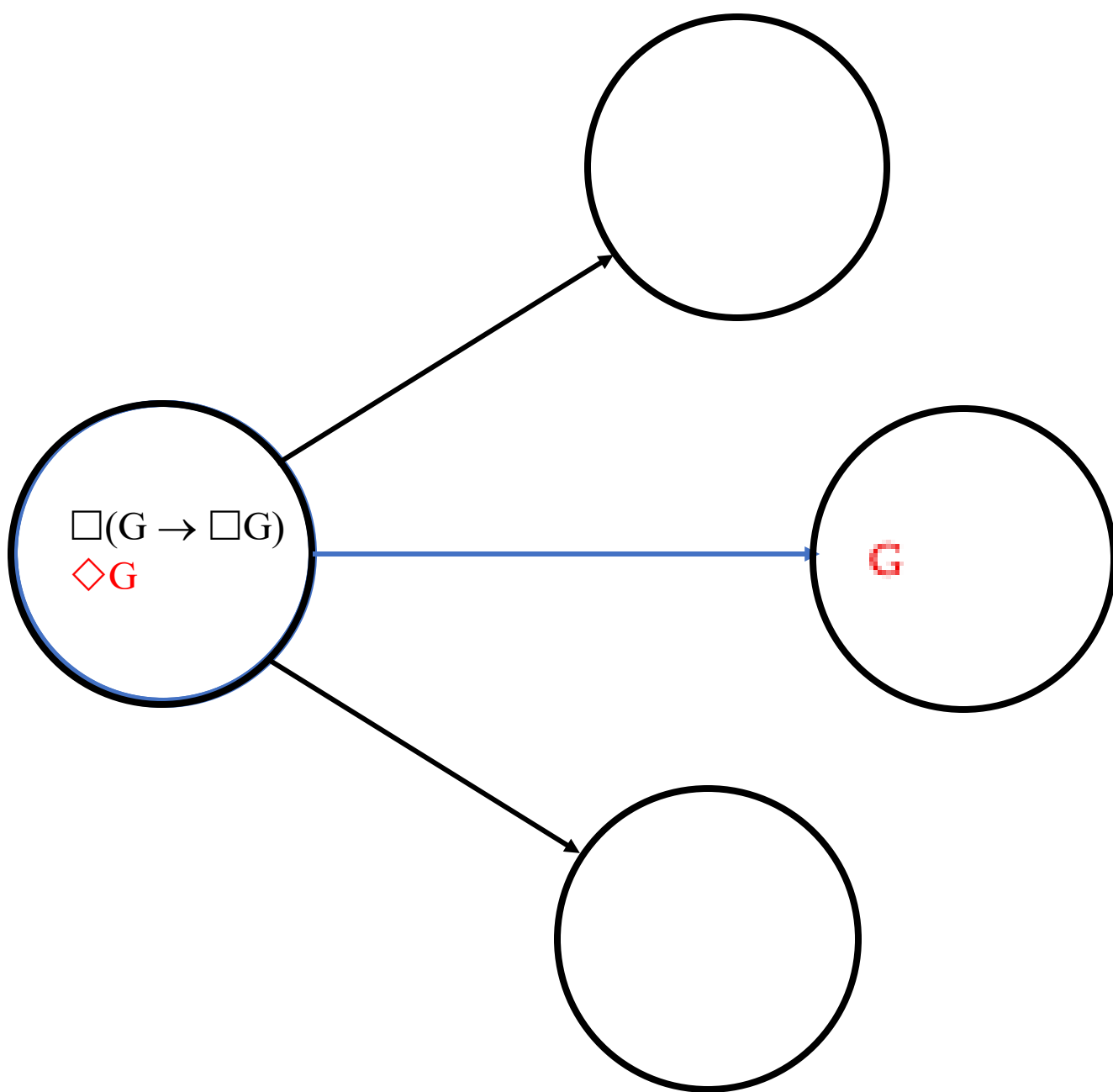
AX. 4 $P(E)$, "Necessary existence is positive"

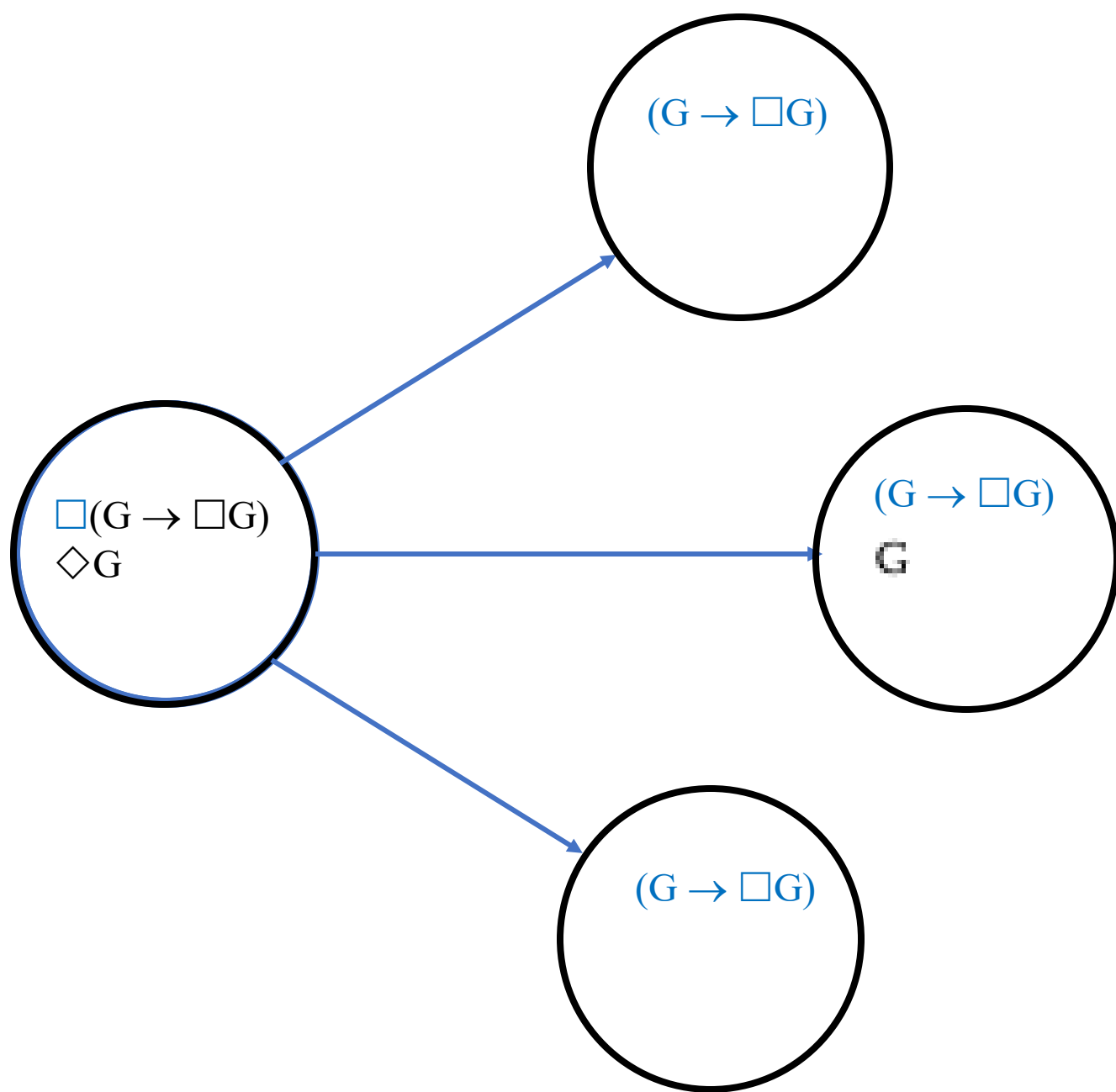
$\therefore \Diamond \Box \exists x Gx \rightarrow \Box \exists x Gx$, S5

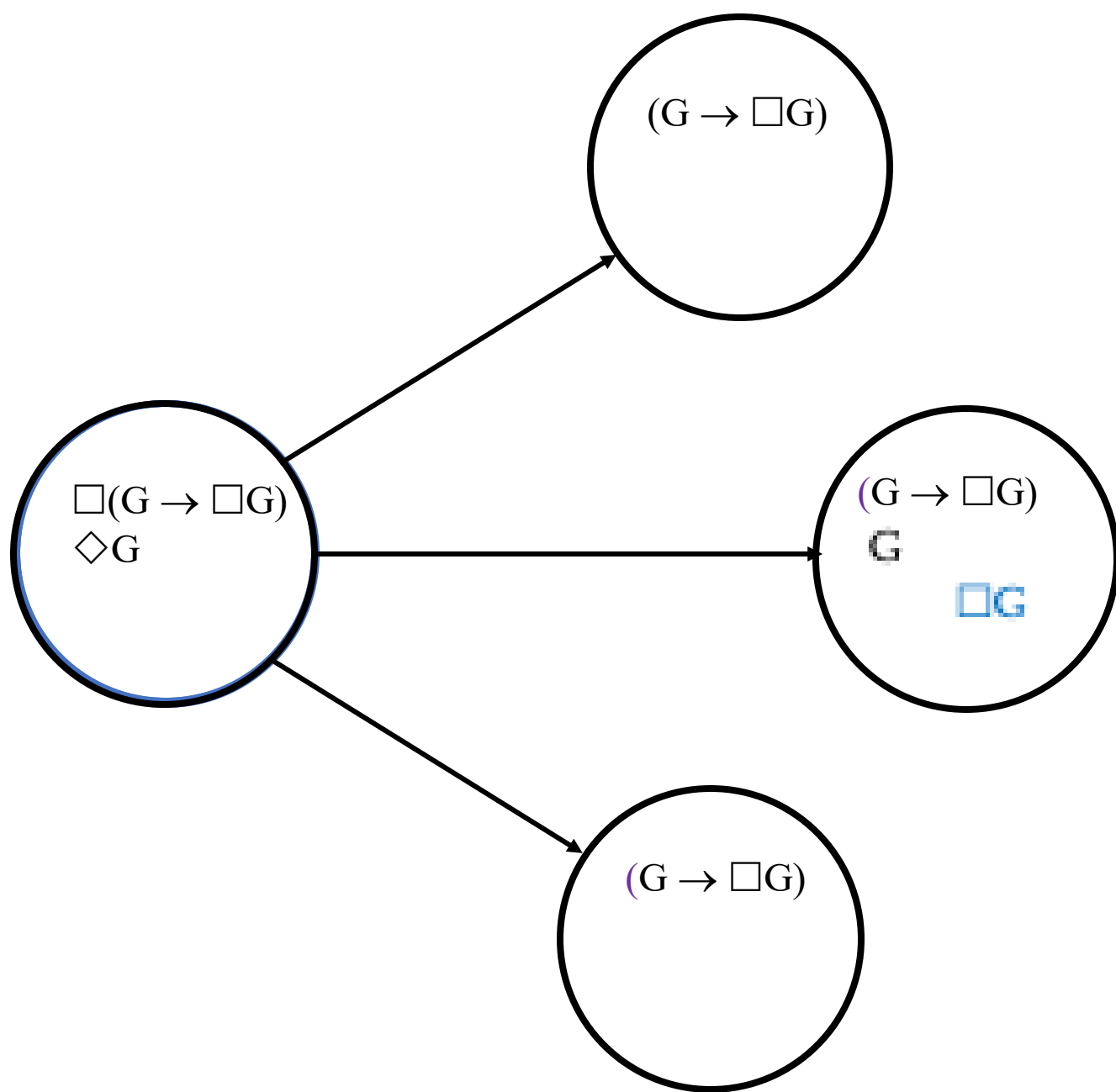
It is necessary that there exists a (unique) god-like being.



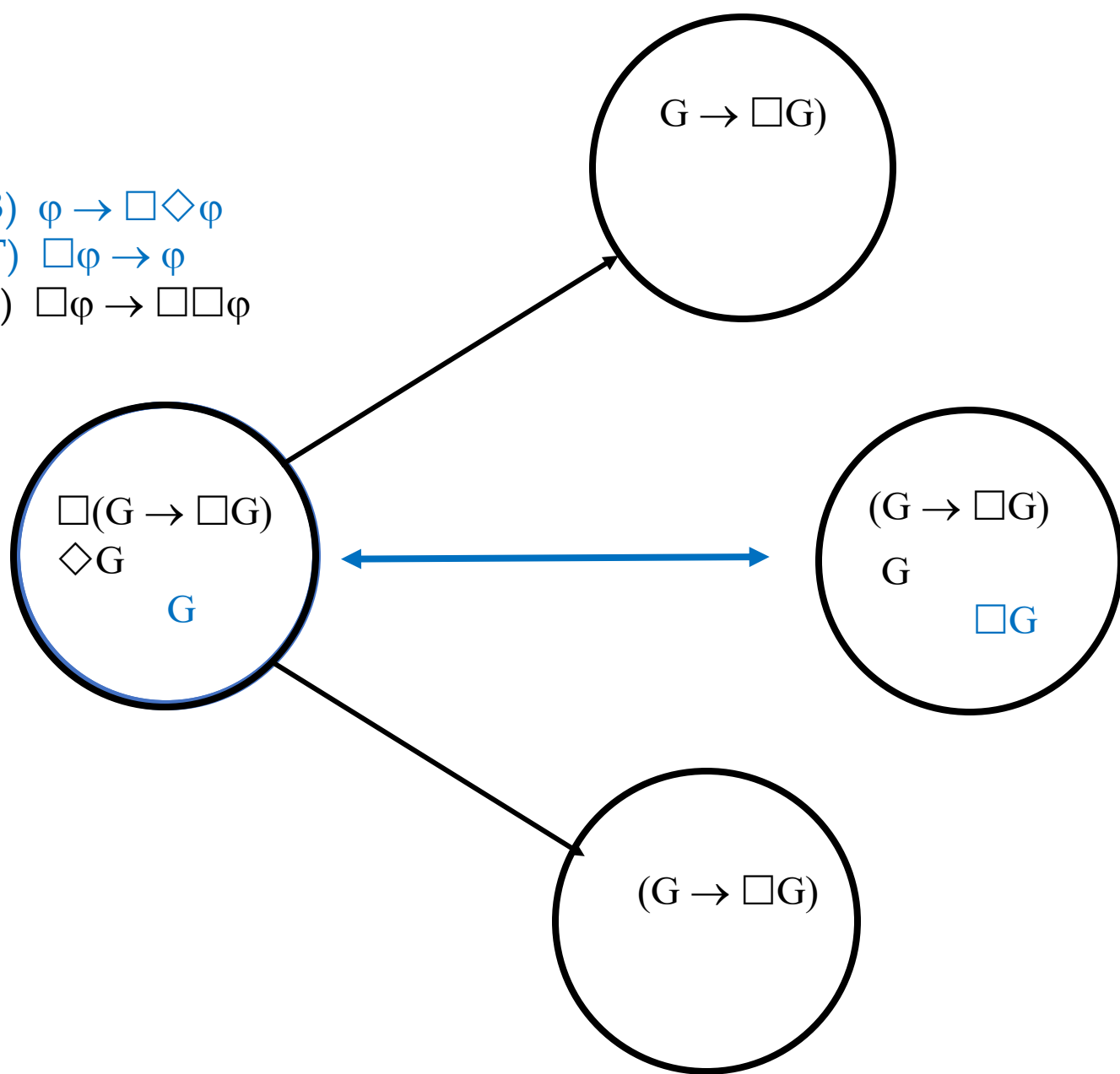






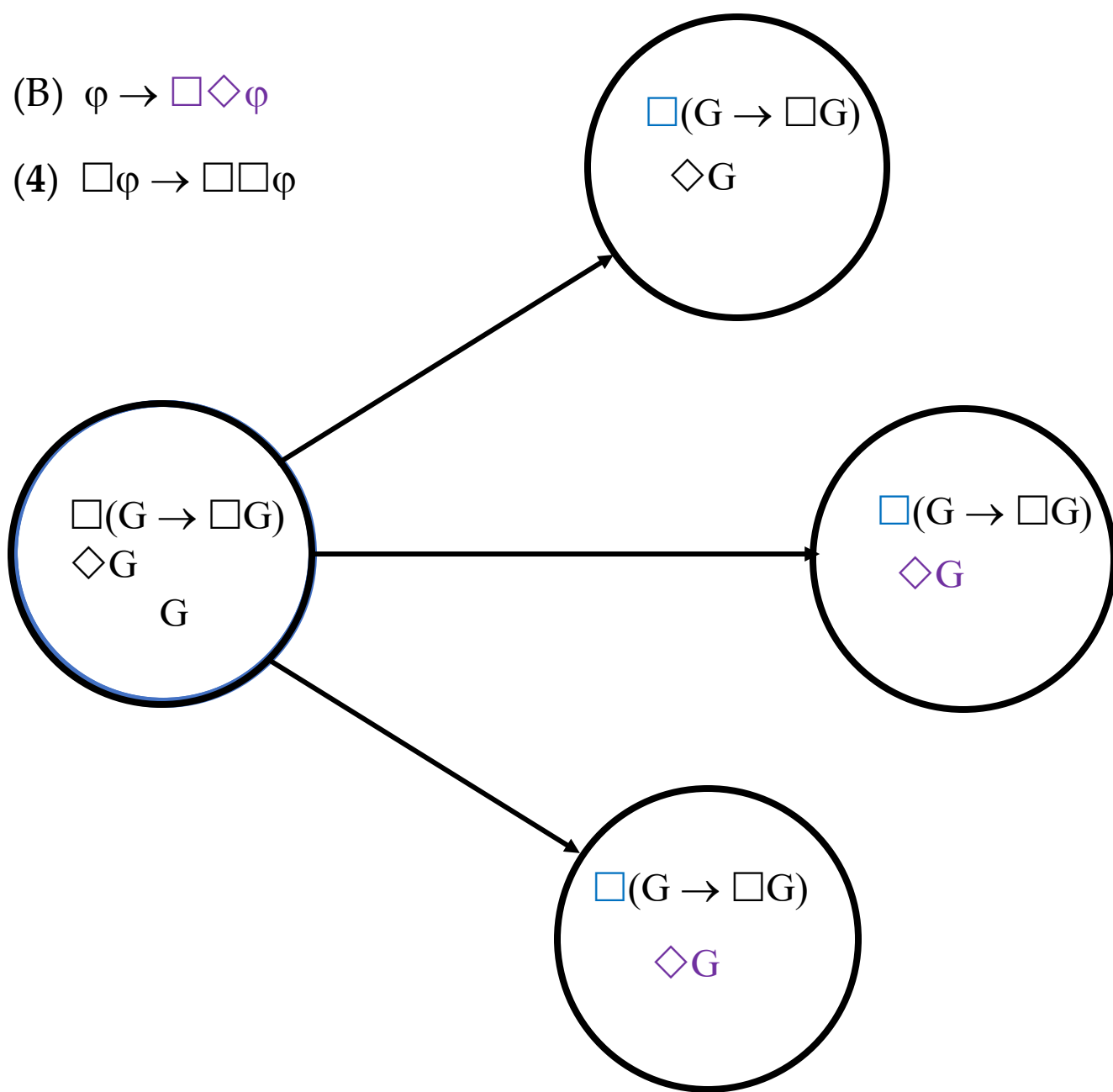


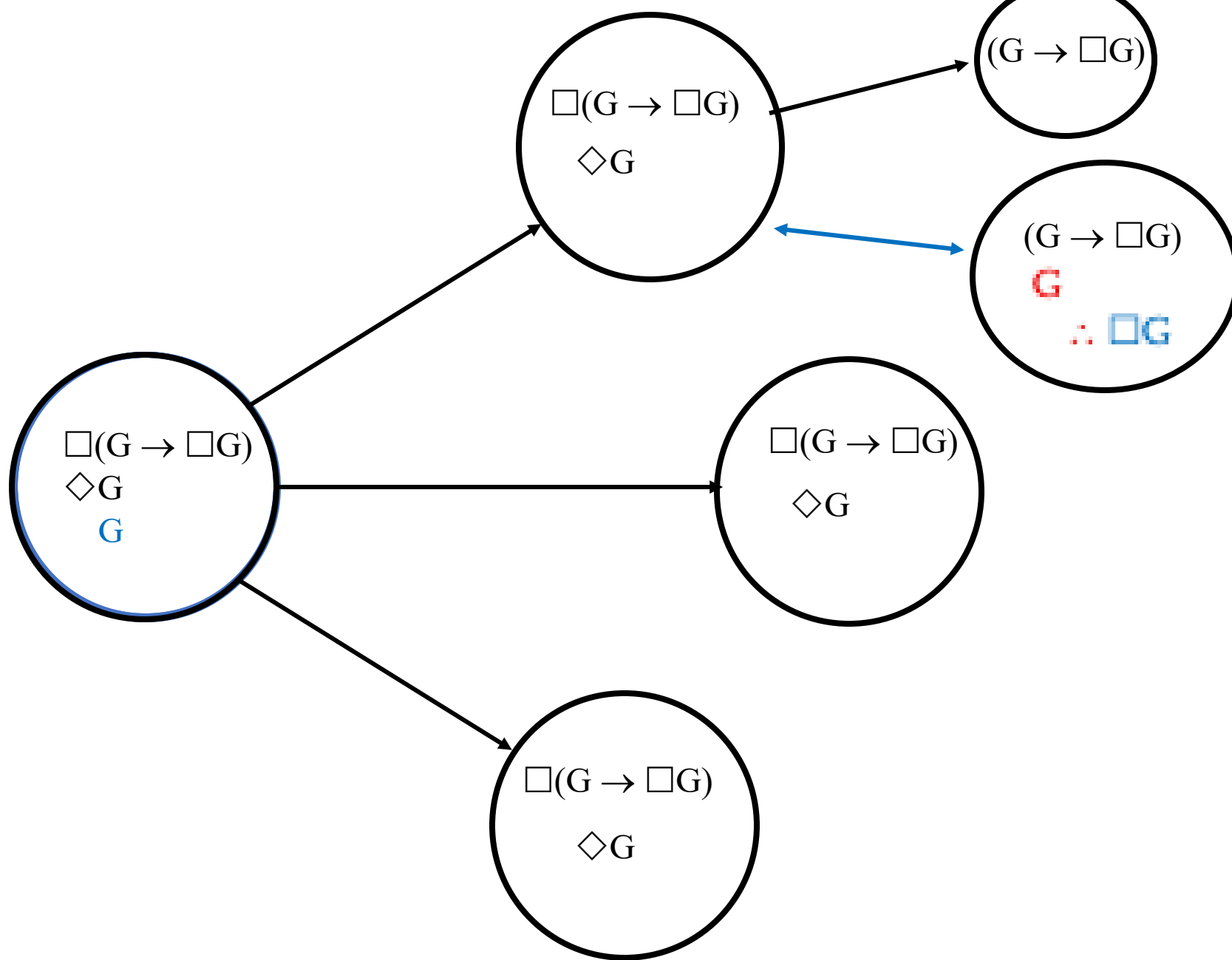
- (B) $\varphi \rightarrow \Box \Diamond \varphi$
- (T) $\Box \varphi \rightarrow \varphi$
- (4) $\Box \varphi \rightarrow \Box \Box \varphi$

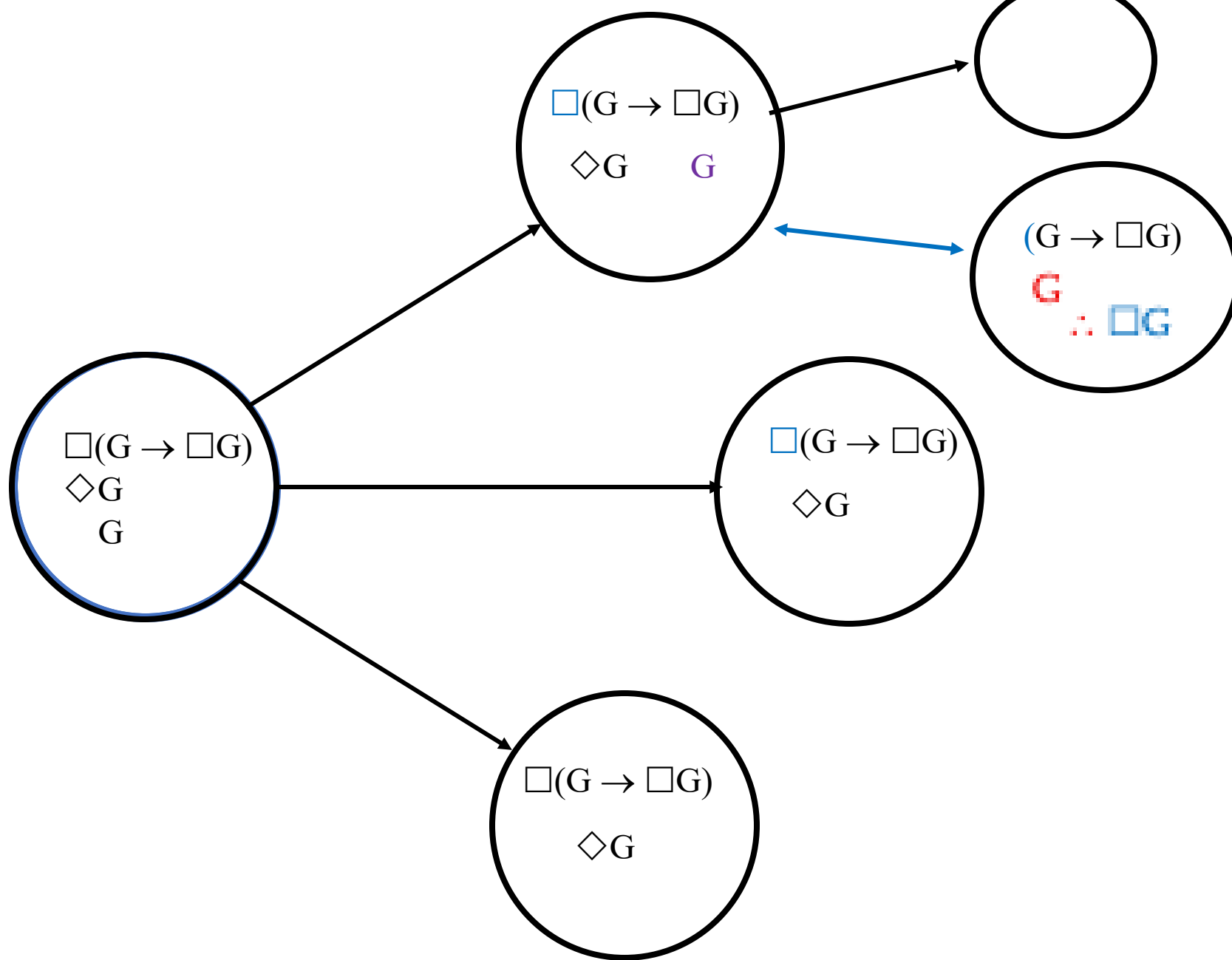


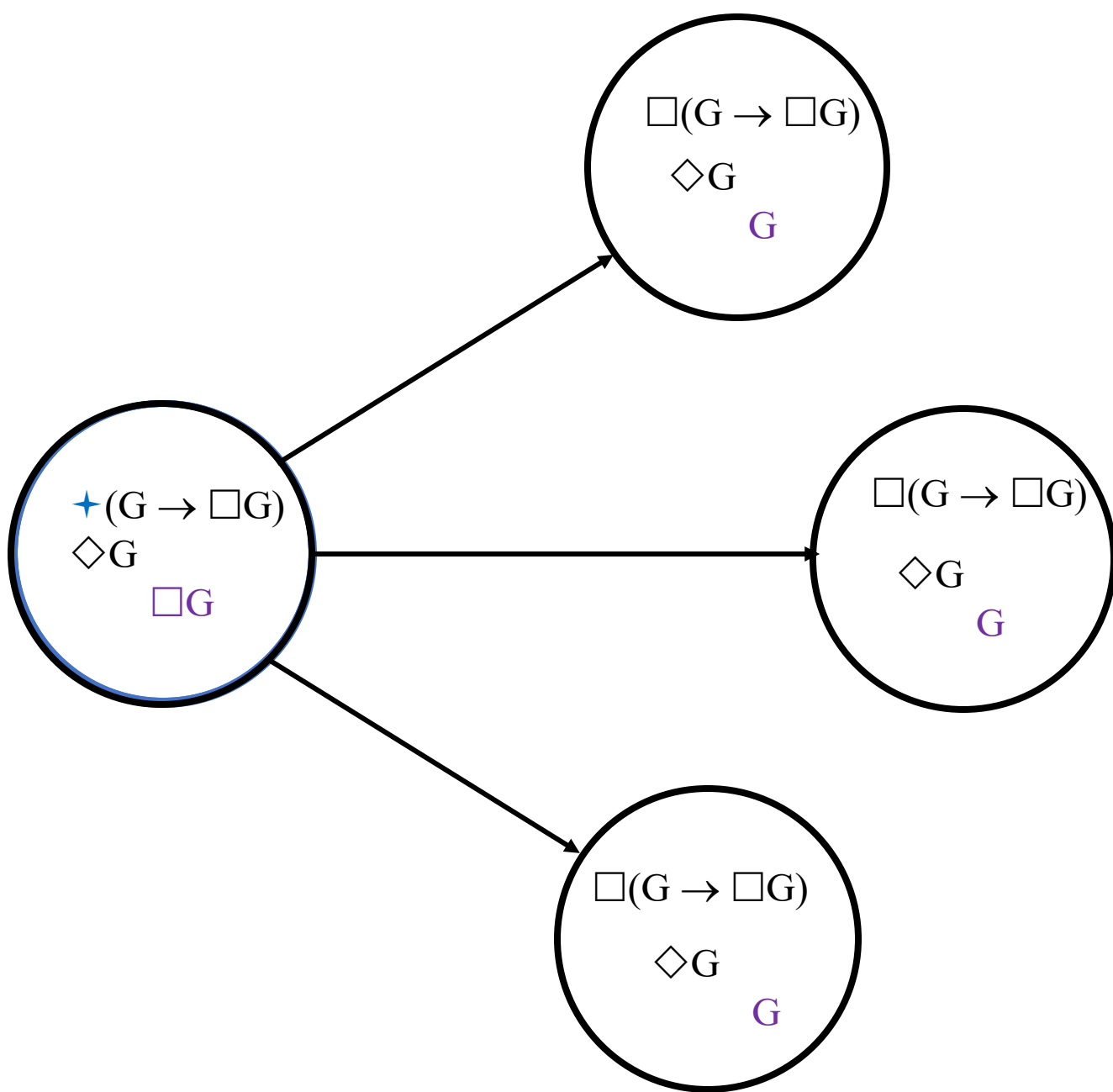
(B) $\varphi \rightarrow \Box \Diamond \varphi$

(4) $\Box \varphi \rightarrow \Box \Box \varphi$









Two Concepts of Proof

Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers

Christoph Benzmüller¹ and Bruno Woltzenlogel Paleo²

Abstract. Kurt Gödel's ontological argument for God's existence has been formalized and automated on a computer with higher-order automated theorem provers. From Gödel's premises, the computer proved: necessarily, there exists God. On the other hand, the theorem provers have also confirmed prominent criticism on Gödel's ontological argument, and they found some new results about it.

The background theory of the work presented here offers a novel perspective towards a *computational theoretical philosophy*.

1 INTRODUCTION

Kurt Gödel proposed an argumentation formalism to prove the existence of God [23, 30]. Attempts to prove the existence (or non-existence) of God by means of abstract, ontological arguments are an old tradition in western philosophy. Before Gödel, several prominent philosophers, including St. Anselm of Canterbury, Descartes and Leibniz, have presented similar arguments. Moreover, there is an impressive body of recent and ongoing work (cf. [31, 19, 18] and the references therein). Ontological arguments, for or against the existence of God, illustrate well an essential aspect of metaphysics: some (necessary) facts for our existing world are deduced by purely a priori, analytical means from some abstract definitions and axioms.

What motivated Gödel as a logician was the question, whether it is possible to deduce the existence of God from a small number of foundational (but debatable) axioms and definitions, with a mathematically precise, formal argumentation chain in a well defined logic.

In theoretical philosophy, formal logical confrontations with such ontological arguments had been so far (mainly) limited to paper and pen. Up to now, the use of computers was prevented, because the logics of the available theorem proving systems were not expressive enough to formalize the abstract concepts adequately. Gödel's proof uses, for example, a complex higher-order modal logic (HOML) to handle concepts such as *possibility* and *necessity* and to support quantification over individuals and properties.

Current works [10, 9] of the first author and Paulson illustrate that many expressive logics, including quantified (multi-)modal logics, can be embedded into the classical higher-order logic (HOL), which can thus be seen as a universal logic [6]. For this universal logic, efficient automated theorem provers have been developed in recent years, and these systems were now employed in our work.

Gödel defines God (see Fig. 1) as a being who possesses all *positive* properties. He does not extensively discuss what positive properties are, but instead he states a few reasonable (but debatable) ax-

A1	Either a property or its negation is positive, but not both: $\forall \phi (P(\neg \phi) \equiv \neg P(\phi))$
A2	A property necessarily implied by a positive property is positive: $\forall \phi \forall \psi ((P(\phi) \wedge \Box \psi \rightarrow \phi(x)) \supset P(\psi))$
T1	Positive properties are possibly exemplified: $\forall \phi (P(\phi) \supset \Diamond \exists x \phi(x))$
D1	A <i>God-like</i> being possesses all positive properties: $G(x) \equiv \forall \phi (P(\phi) \supset \phi(x))$
A3	The property of being God-like is positive: $P(G)$
C	Possibly, God exists: $\Diamond \exists x G(x)$
A4	Positive properties are necessarily positive: $\forall \phi (P(\phi) \supset \Box P(\phi))$
D2	An <i>essence</i> of an individual is a property possessed by it and necessarily implying any of its properties: $\emptyset \text{ ess. } x \equiv \phi(x) \wedge \forall \psi (\phi(x) \supset \Box \forall y (\phi(y) \supset \psi(y)))$
T2	Being God-like is an essence of any God-like being: $\forall x (G(x) \supset G \text{ ess. } x)$
D3	<i>Necessary existence</i> of an individ. is the necessary exemplification of all its essences: $NE(x) \equiv \forall \phi (\phi \text{ ess. } x \supset \Box \exists y (\phi(y)))$
A5	Necessary existence is a positive property: $P(NE)$
T3	Necessarily, God exists: $\Box \exists x G(x)$

Figure 1. Scott's version of Gödel's ontological argument [30].

ioms that they should satisfy. Various slightly different versions of axioms and definitions have been considered by Gödel and by several philosophers who commented on his proof (cf. [31, 3, 2, 19, 1, 18]).

The overall idea of Gödel's proof is in the tradition of Anselm's argument, who defined God as some entity of which nothing greater can be conceived. Anselm argued that existence in the actual world would make such an assumed being even greater; hence, by definition God must exist. Gödel's ontological argument is clearly related to this reasoning pattern. However, it also tries to fix some fundamental weaknesses in Anselm's work. For example, Gödel explicitly proves that God's existence is possible, which has been a basic assumption of Anselm. Because of this, Anselm's argument has been criticized as incomplete by Leibniz. Leibniz instead claimed that the assumption should be derivable from the definition of God as a perfect being and from the notion of perfection. Gödel's proof addresses this critique, and it also addresses the critique of others, including Kant's objection that existence should not be treated as a predicate. On the other hand, Gödel's work still leaves room for criticism, in particular, his axioms are so strong that they imply *modal collapse*, that is, a situation where contingent truths and necessary truths coincide. More information on the philosophical debate on Gödel's proof is provided in [31].

Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-16)

The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics

Christoph Benzmüller^{*}
Freie Universität Berlin & Stanford University
c.benzmueller@gmail.com

Bruno Woltzenlogel Paleo
Australian National University
bruno.wp@gmail.com

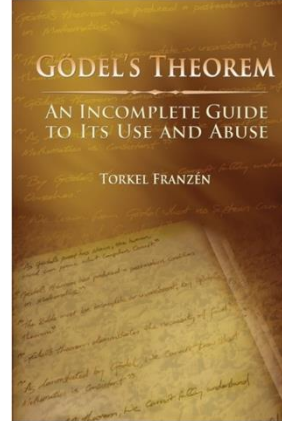
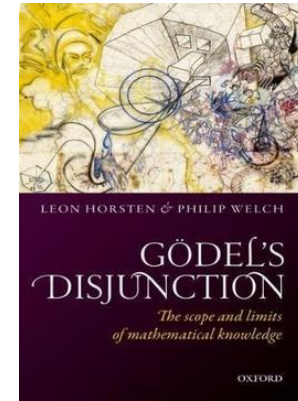
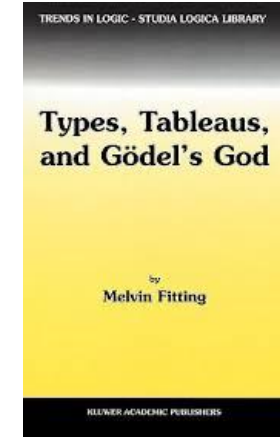
Abstract

This paper discusses the discovery of the inconsistency in Gödel's ontological argument as a success story for artificial intelligence. Despite the popularity of the argument since the appearance of Gödel's manuscript in the early 1970's, the inconsistency of the axioms used in the argument remained unnoticed until 2013, when it was detected automatically by the higher-order theorem prover

on the proof [Führmann, 2016].

The in-depth analysis presented here substantially extends previous computer-assisted studies of Gödel's ontological argument. Similarly to the related work [Benzmüller and Woltzenlogel-Paleo, 2013a; 2014] the analysis has been conducted with automated theorem provers for classical higher-order logic (HOL; cf. [Andrews, 2014] and the references therein), even though Gödel's proof is actually formulated in higher-order modal logic (HOML; cf. [Muskens, 2006])

6.1.8. My incompleteness theorem makes I likely that mind is not mechanical, or else mind cannot understand its own mechanism. If my result is taken together with the rationalistic attitude which Hilbert had and which was not refuted by my results, then [we can infer] the sharp result that mind is not mechanical. This is so because: if the mind were a machine, there would, contrary to the rationalistic attitude, exist number-theoretical questions undecidable for the human mind. (Wang [1996], 6.1.8, 186-7.



What makes a Rational Interstellar Elegant proof?

- (1) **Formal** syntactical derivation vs. **Rational** intuitive proof
 - (A) **HILBERT PROGRAM**: formalist mechanistic “decision procedures”
 - (B) **GÖDEL PROGRAM**: intuitive creative “search for new axioms”
 - (C) Gödel Disjunction: *Modus Tollendo Ponens*
- (2) Conceptual map or **“constellation of ideas”**
 - (A) *axiomatized* “mathematically”
 - (B) *defined* with “informal rigour”
 - (C) *confirmed* with a network of derivations
- (3) **Elegance**, beauty, “aha!” factor
 - (A) *memorability*—rational *reconstructibility*
 - (4) *compactness*—minimal *width* & maximal *depth*
 - (5) *phenomenology*—succinct, symmetrical, delight or surprise

¹ Freie Universität Berlin, Germany, email: c.benzmueller@fu-berlin.de; this author has been supported by the German National Research Foundation (DFG) under grants BE 2501/9-1 and BE 2501/11-1.

² Technical University Vienna, Austria, email: bruno@logic.at

or a philosopher who awake to find he was a logician ...

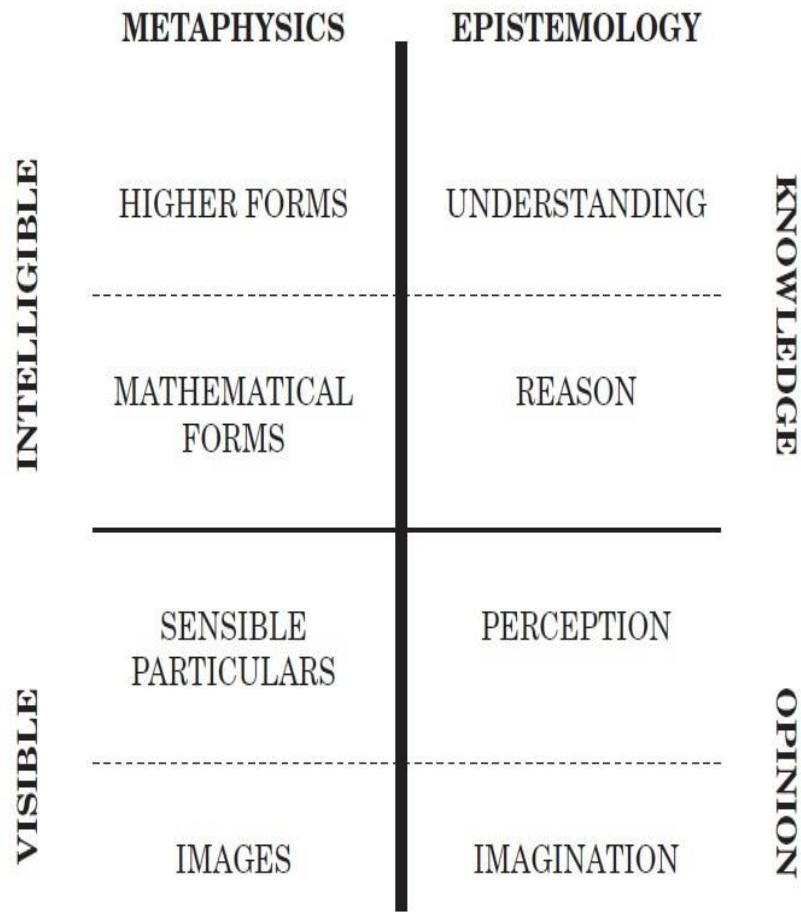
1. "The world is rational."
2. "There is incomparably more knowable *a priori* than is currently known."
3. Logic, mathematics, physics reveal the structure of the world.
4. Mechanistic formal systems cannot be the cause of rationality because mathematical logic can prove their incompleteness.
5. Either the mind transcends formal systems or there are absolutely unsolvable mathematical problems.
6. If relativity reveals the possible structures of time, then time is not objectively real.
7. The only sufficient reason (or "First Cause") of the objective existence and beautiful of mathematics is an infinite mind with all perfections.



"Die Welt ist vernünftig."

— KURT GÖDEL, "My Philosophical Outlook" (c. 1960)

Plato's Divided Line



- Plato's Form of the Good illuminates Rationality, Beauty, and Goodness.
- Even if we do not see the Sun directly we know it exists because by its light we see all else.
- Juliet Kennedy [2008]: "for Gödel ... Platonism is a thesis about the pursuit of understanding; not for its own sake, but because ... by pursuing understanding we exercise our connection to the good."

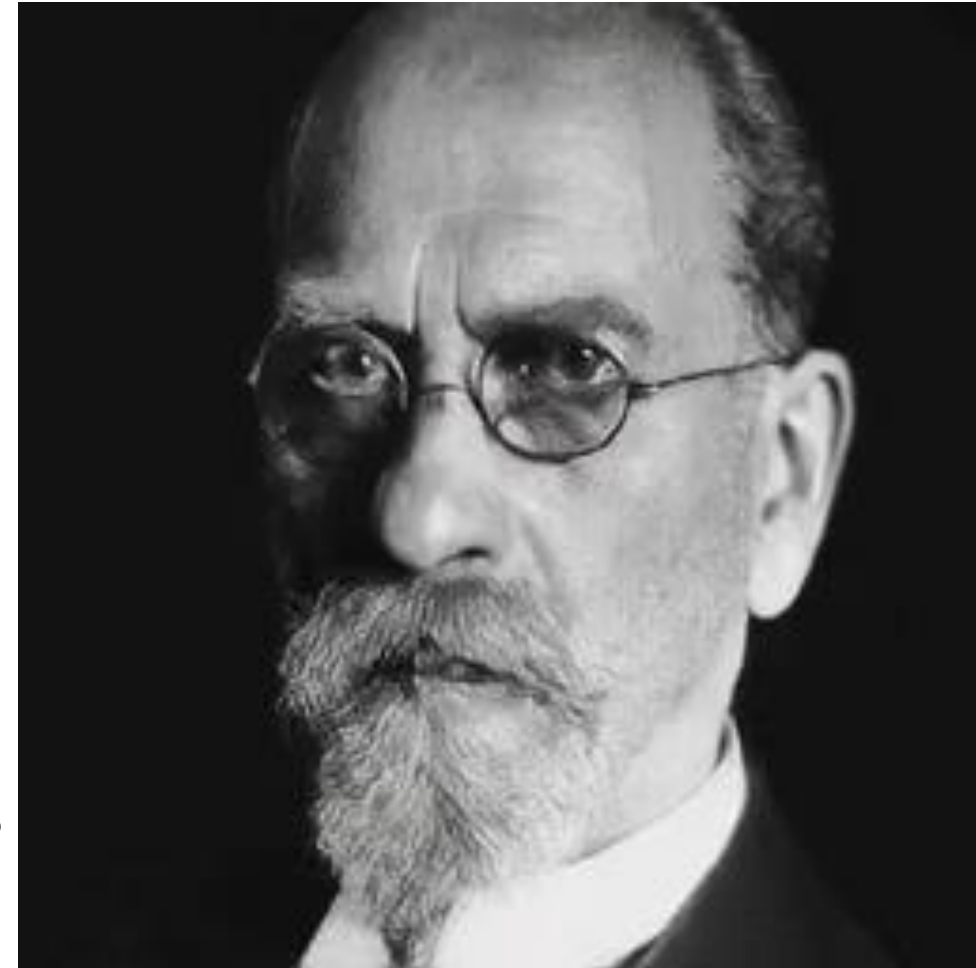
“I myself have never had such an experience. For me, there is no absolute knowledge”

... in his turn to Husserl's philosophy in 1959, but apparently in vain:

“At some time between 1906 and 1910 Husserl had a psychological crisis. He doubted whether he had accomplished anything and his wife was very sick. At some point in this period, everything suddenly became clear to Husserl, and he did arrive at some absolute knowledge. But one cannot transfer absolute knowledge to somebody else; therefore, one cannot publish it. A lecture on the nature of time also came during this period when Husserl's experience of seeing absolute knowledge took place. I myself have never had such an experience. For me, there is no absolute knowledge: everything goes only by probability. Both Descartes and Schelling explicitly reported an experience of sudden illumination when they began to see everything in a different light.”

Why did this “sudden illumination” elude Gödel's logical mind? Husserl described this phenomenological turn in a letter written several decades later to Sister Edelgundis (I am indebted to Anthony Steinbach for this reference):

“I also want what the churches want: to lead humanity to the *Aeternitas*. My task is to try to do this through philosophy. Everything that I have written up to now is only preparatory; it is only a development [*Aufstellen*] of methods. In the course of one's life, one unfortunately does not arrive at the core, at what is essential. It is so important for philosophy to be led out of liberalism and rationalism, and to be led once more to what is essential, to truth. The question concerning ultimate reality, truth, must be the object of every true philosophy. This is my life's work.”



*Idolatry, the desire for absolute good
and beauty leads one "out of the cave"*



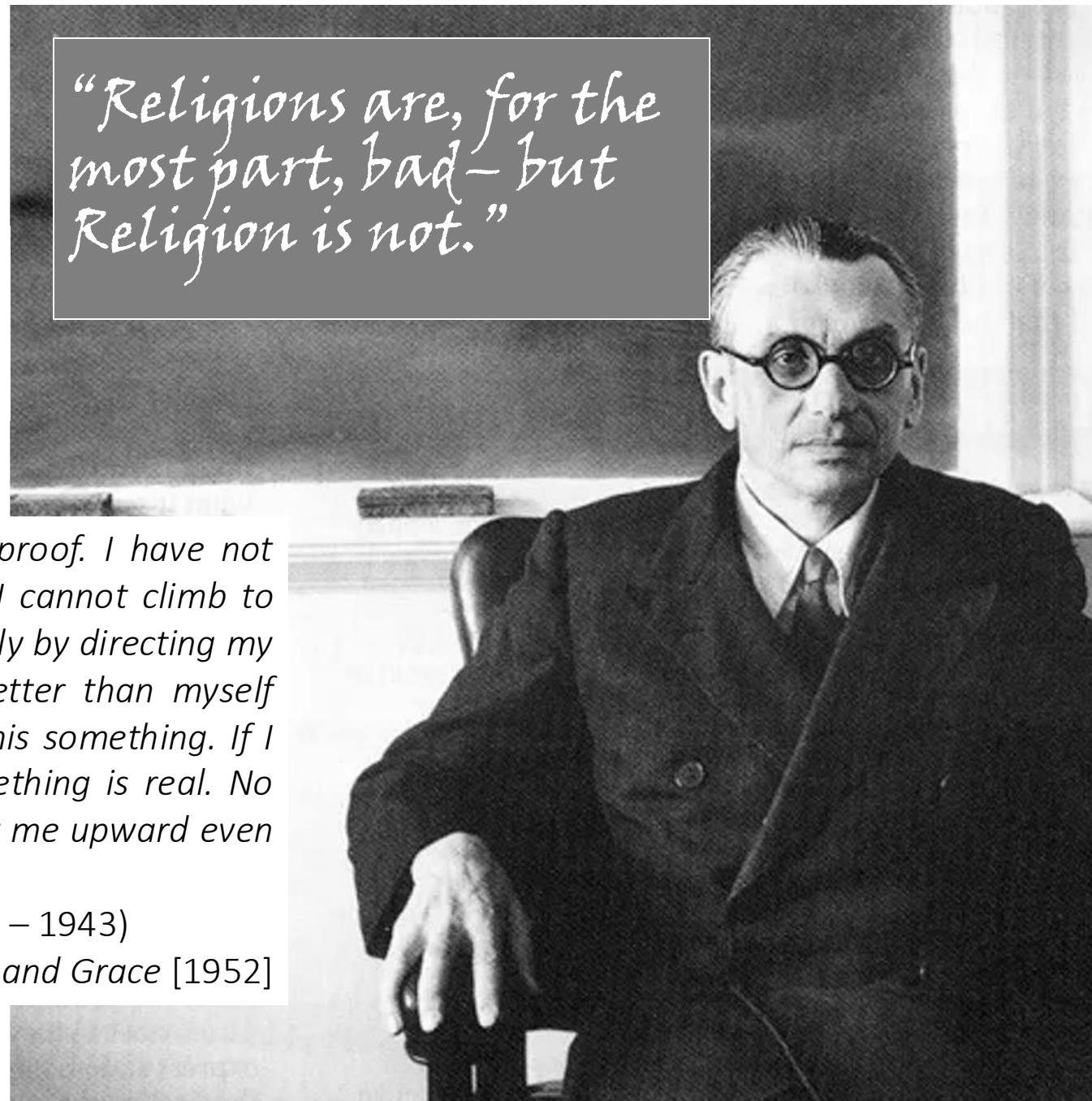
"God exists since mathematics is consistent, and the Devil exists since we cannot prove it."
—ANDRÉ WEIL (6 May 1906 – 6 Aug. 1998)



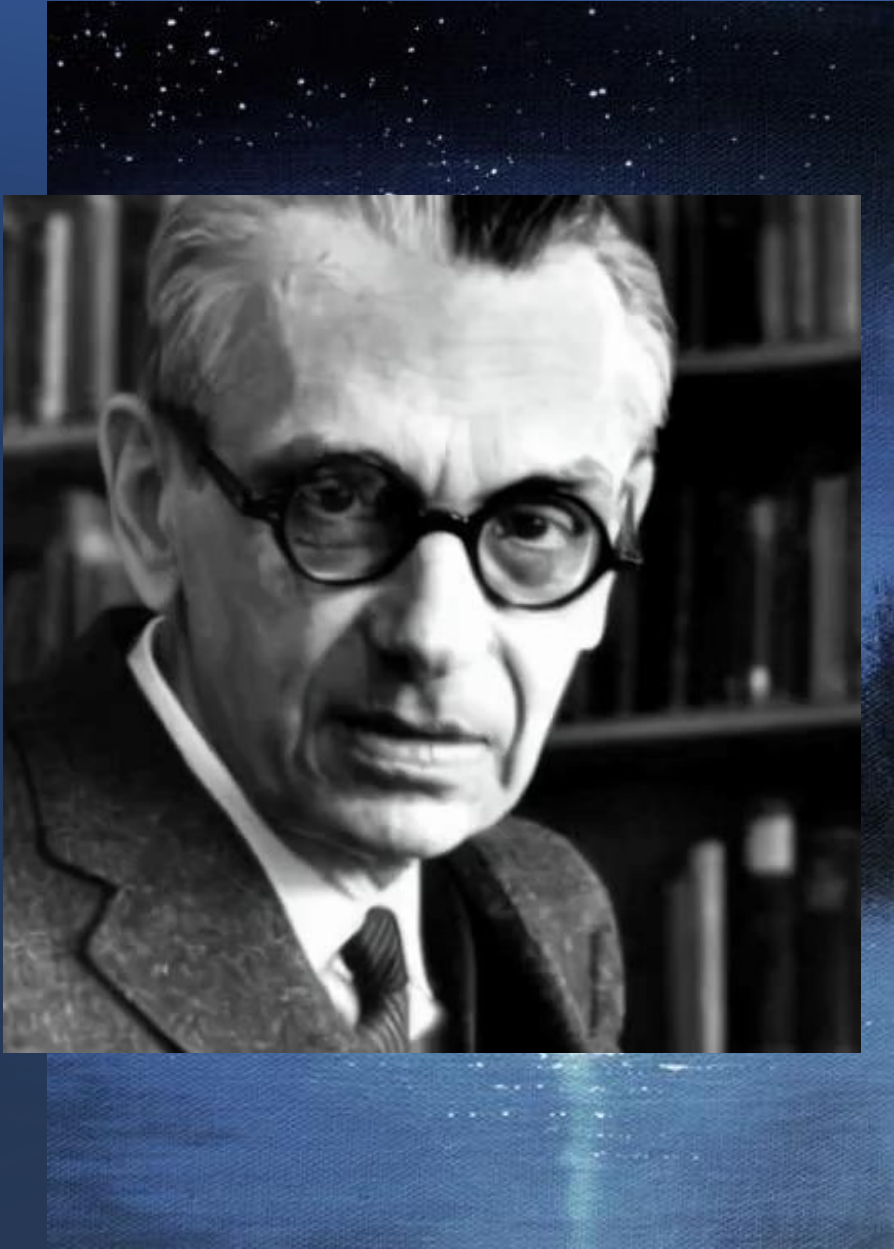
"An experimental ontological proof. I have not the principle of rising in me. I cannot climb to heaven through the air. It is only by directing my thought toward something better than myself that I am drawn upward by this something. If I am really raised up, this something is real. No imaginary perfection can draw me upward even by the fraction of an inch."

— SIMONE WEIL (1909 – 1943)

Gravity and Grace [1952]



"Religions are, for the most part, bad – but Religion is not."



“In principle, we can know all mathematics. It is given to us in its entirety and does not change—unlike the Milky Way. That part of it of which we have a perfect view seems beautiful suggesting harmony; that is that all the parts fit together although we see fragments of them only.

Inductive inference is not like mathematical reasoning: it is based on equality or uniformity. But mathematics is applied to the real world and has proved fruitful. This suggests that the mathematical and empirical parts are in harmony and that the real world is also beautiful.

Otherwise, mathematics would be just an ornament and the real world would be like an ugly body in beautiful clothing.

(Wang [1996], p4.4.18m . 151)

Gödel's Ontological Dream

Mathematical Realism, Temporal Unrealism, and Gödel's God



Professor Gary Mar
Stony Brook University
Stony Brook, New York, USA



December 5, 2025 [Gödel Freundkreis](#)
Bamberg University, Germany

Church died in Hudson on August 11, 1995. Following services at the the University Chapel at Princeton University, he was buried in the Princeton Cemetery. In addition to his son, he is survived by two daughters, Mary Ann Addison and Mildred Dandridge, and by eight grandchildren and two great-grandchildren.

His "academic offspring" are of course his doctoral students. They form a distinguished list indeed:

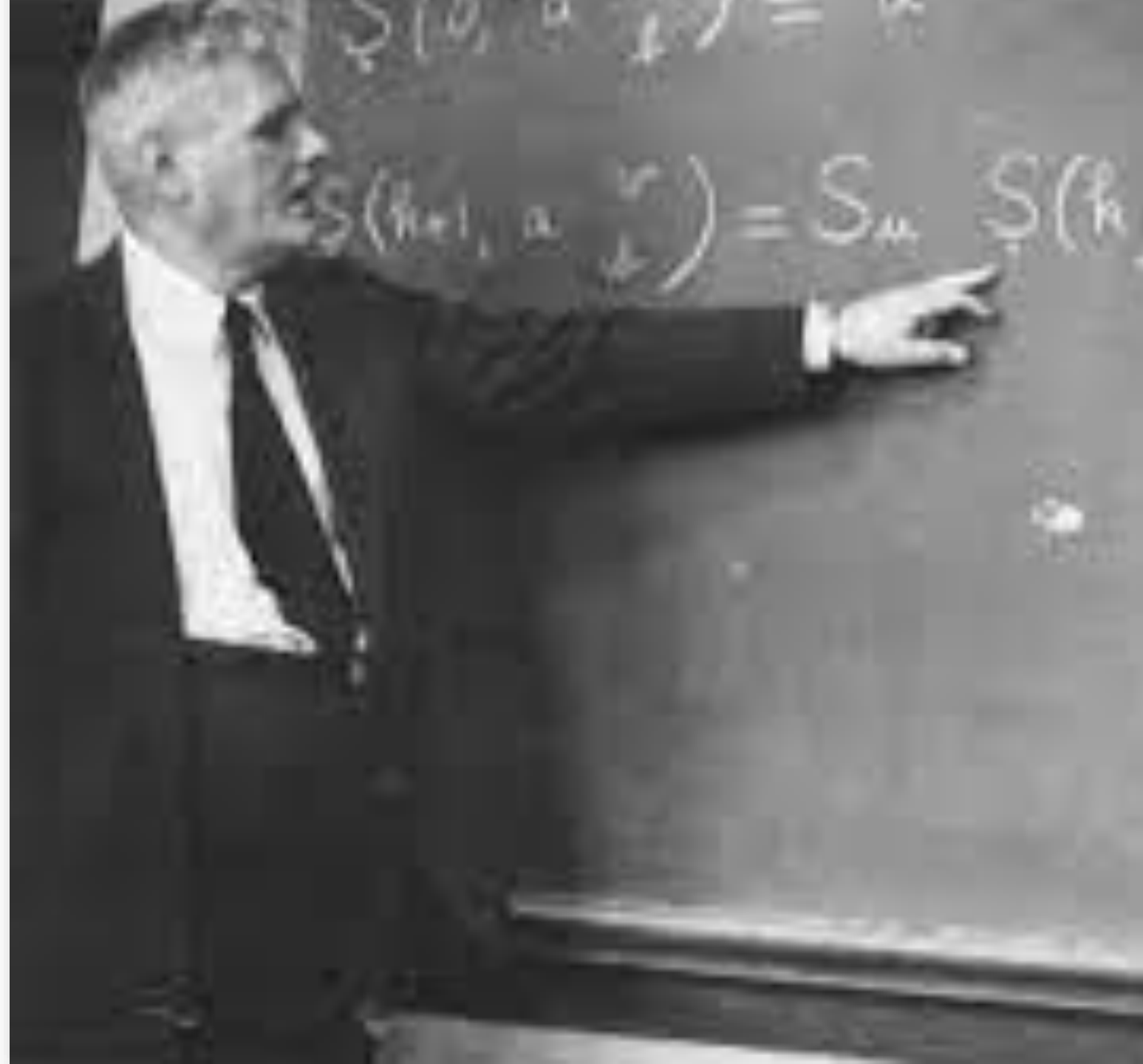
1931 Alfred L. Foster	1959 Aubert Daigneault
1934 Stephen C. Kleene	1959 Raymond Smullyan
1934 J. Barkley Rosser	1960 Robert W. Ritchie
1938 Alan M. Turing	1961 James R. Guard
1944 Enrique Bustamante-Llaca	1962 James Bennett
1947 Leon Henkin	1962 Robert O. Winder
1949 John G. Kemeny	1963 Wayne H. Richter
1950 Martin D. Davis	1963 Gustav B. Hensel
1951 Maurice L'Abbé	1964 Peter Andrews
1952 William W. Boone	1964 William B. Easton
1952 Hartley Rogers, Jr.	1965 Joel W. Robbin
1955 Norman Shapiro	1967 Donald J. Collins
1956 T. Thacher Robinson	1976 Richard J. (Isaac) Malitz
1957 Michael O. Rabin	1977 C. Anthony Anderson
1958 Dana Scott	1985 Gary R. Mar
1959 Simon Kochen	

Church was elected to the National Academy of Sciences in 1978. He was a corresponding member of the British Academy, and a member of the American Academy of Arts and Sciences. He received honorary doctorates from Case Western Reserve (1969), Princeton (1985), and the State University of New York at Buffalo (1990).

An edition of Church's collected papers is planned. It will include a complete bibliography of his publications: not only the books and the research articles, but the numerous reviews (some of which were very influential) and his expository encyclopedia articles.

H. B. ENDERTON

(1936 – 2010)



“To the extent that one wishes to refer to Gödel’s Platonism, his remarks here may say something important about his notion of it, something which has perhaps been forgotten in the present-day literature on Gödel’s use of the term. Namely, for Gödel, at the end of the day, literally, Platonism is a thesis about the pursuit of understanding; not for its own sake, but because—and this is certainly the very heart of Platonism—by pursuing understanding we exercise our connection to the good. Our intellectual aspirations and intellectual achievements are nothing other than moral aspirations and moral achievements.”



Juliette Kennedy [2008] “Two Moments in the Philosophical Life of Kurt Gödel: Completeness and the Separation of the First-and Higher-Order Logic, and Reflections on Moral Platonism”.

"Materialism is false."



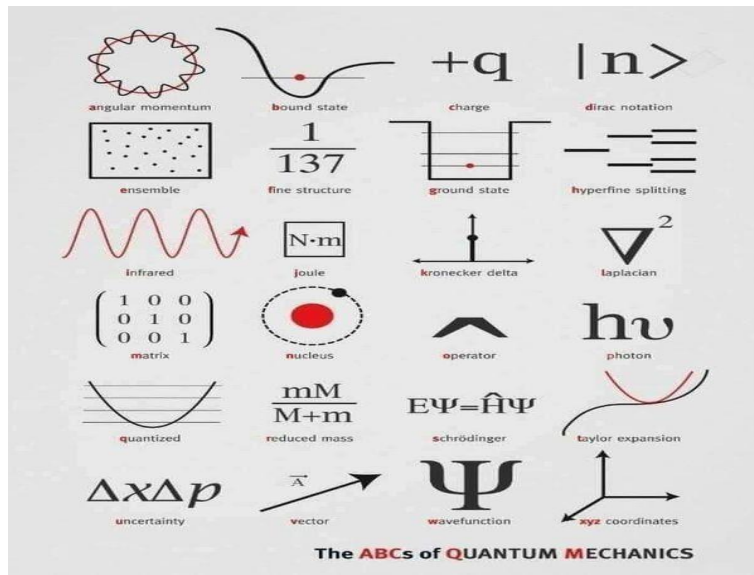
One of Gödel's 14 propositions is that "materialism is false". Gödel's conceptual realism is inconsistent with materialism, but so is Turing's affirmation of naturalism. In March 1954 shortly before his death, Turing sent Robin Gandy, his sole Ph.D. student, four postcards entitled "*Messages from the Unseen World*" (alluding to Eddington's 1929 lecture "*Science and the Unseen World*").

At the time Turing was investigating how nature could produce non-computational outcomes, e.g., how the Zebra's stripes could be generated by differential equations. On one of these "Messages from the Unseen World", Turing wrote:

"I'm trying to invent a new Quantum Mechanics, but it won't really work."

The world will never know what Turing might have found, for three months later on June 7th, Turing died with a cyanide-laced apple by his bedside.

Turing was committed to *naturalism*, not necessarily materialism. *Naturalism*, or the commitment to the fundamental entities of the currently best scientific theory of the physical world, implies "action at a distance", which is inconsistent with materialism. Therefore, if either platonism or computationalism is true, then materialism is false.

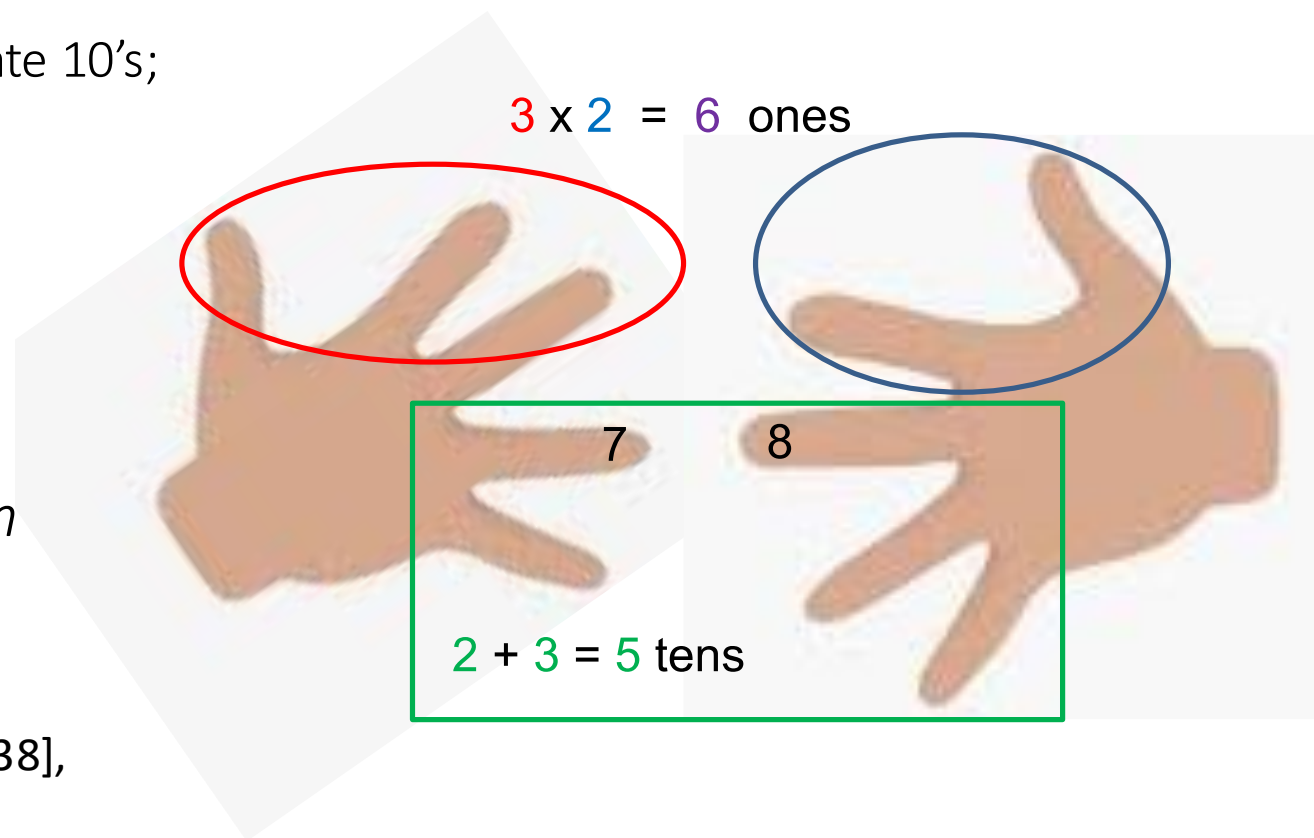


Turing's [1938] Ph.D. Thesis: "Ingenuity & Intuition"

1. Add fingers touching and below to calculate 10's;
2. Multiply fingers above to calculate 1's.

"Mathematical reasoning may be regarded rather schematically as the exercise of a combination of two faculties*, which we may call *intuition* and *ingenuity*."

Turing, "Systems of Logic Based on Ordinals" [1938],
Section 11: [Cooper & Van Leeuwen [2013], 38]



$$7 \times 8 = 50 + 6 = 56$$

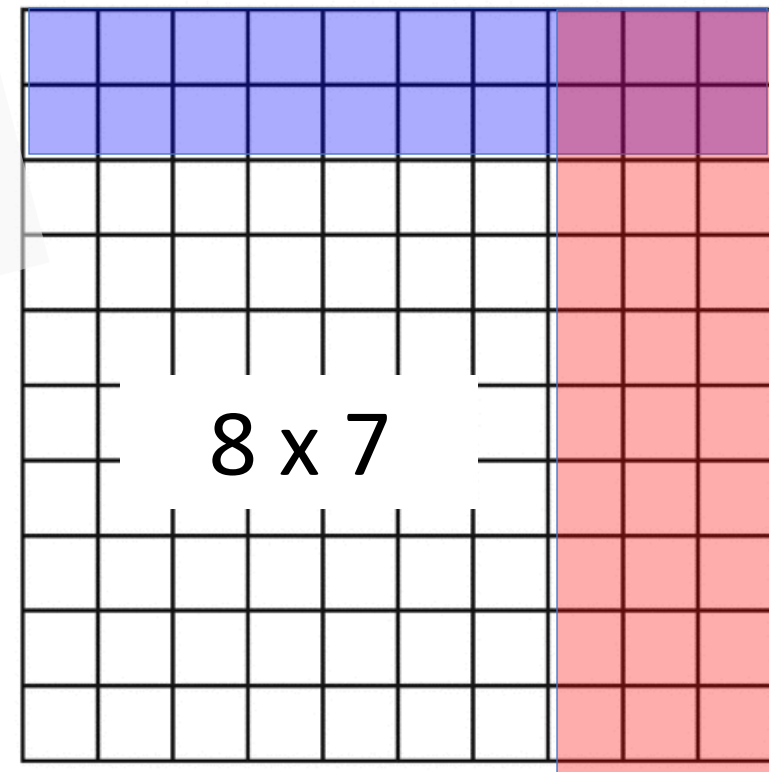
,,, through suitable arrangements of propositions, and perhaps geometrical figures or drawings. It is intended that when these are really well arranged the validity of the intuitive steps which are required cannot seriously be doubted.”

[Cooper & Van Leeuwen [2013], 39]

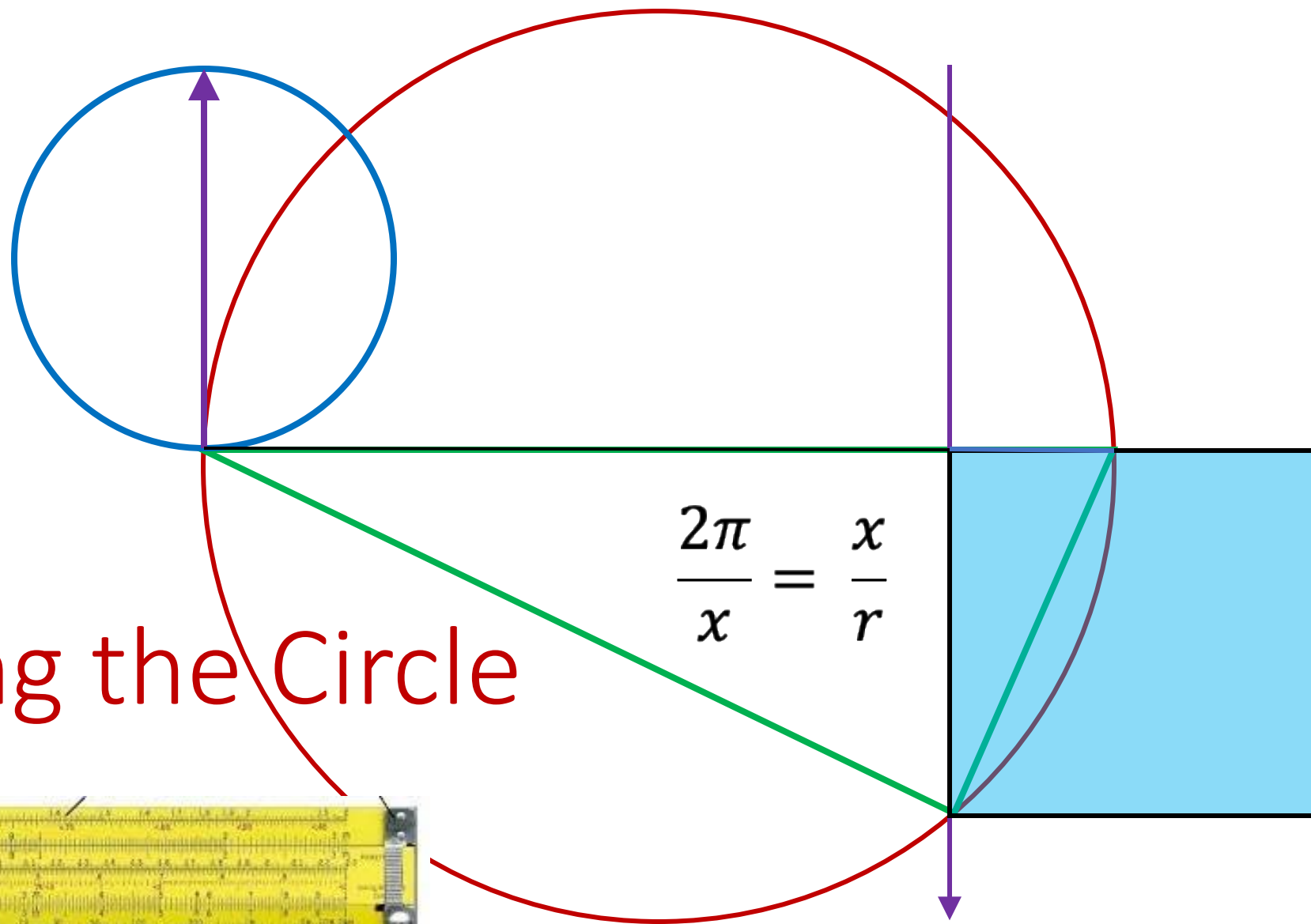
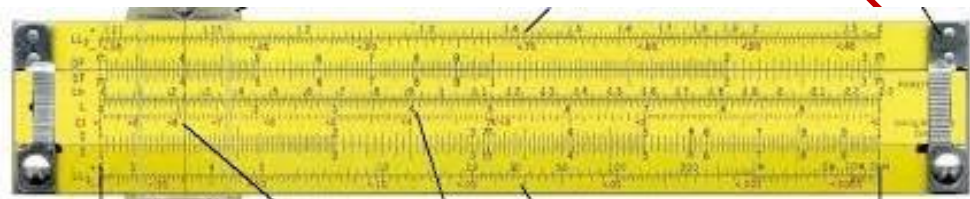
$$\boxed{8 \times 7} = 100 - [(10 - 8) + (10 - 7)] \times 10 + (2 \times 3)$$

$$\begin{aligned} [(7 - 5) + (8 - 5)] \times 10 &= (2 + 3) \times 10 = 50 \\ (10 - 7) \times (10 - 8) &= 2 \times 3 = \underline{6} \\ &56 \end{aligned}$$

$$\begin{aligned} [(X - 5) + (Y - 5)] \times 10 &= -100 + 10X + 10Y \\ (10 - X) \times (10 - Y) &= \underline{100 - 10X - 10Y + XY} \\ &XY \end{aligned}$$



Squaring the Circle



Tarski Platonism and Pathologizing the Paradoxes



Amicus Plato, sed magis amica veritas
[Plato is a friend, but it is right to prefer truth].

:

Plato sed magis inimica falsitas
[Plato is an enemy, but falsehood is a greater enemy].



Tarski would have us see the meta-mathematical pearls of Goedel, Church, Tarski, Turing as merely the “disease of the oyster”—instead of seeking pearls of infinite complexity, chaos, and fractal beauty.

Tarski's concession to Platonism; *Pathologizing* the Paradoxes



Amicus Plato, sed magis amica veritas
[Plato is a friend, but it is right to prefer truth].

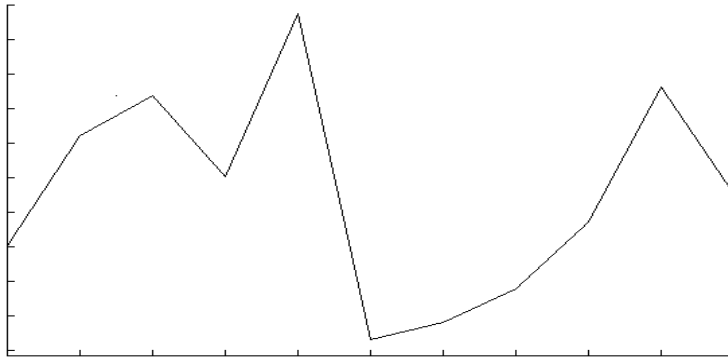
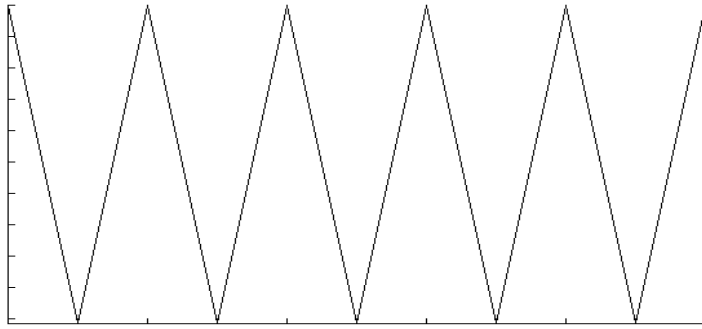
:

Plato sed magis inimica falsitas
[Plato is an enemy, but falsehood is a greater enemy].



Personally, as a logician, I could not reconcile myself with antinomies as permanent elements of our system of knowledge. *The appearance of antinomy is for me a symptom of disease.*

Dynamical Semantics: a computational approach to paradoxes



The boxed sentence is not true.

1. The boxed sentence = “The boxed sentence is not true.”
2. “The boxed sentence is not true” is true if and only if the boxed sentence is not true.
3. Therefore, the boxed sentence is true if and only if the boxed sentence is not true.

ITERATIVE DYNAMICAL SYSTEMS

RESCHER’S VVP SCHEMA

LUKSIEWCIZIAN INFINITE-VALUED LOGIC

- SELF-REFERENCE
- TARSKI’S (T) SCHEMA
- ARISTOTELIAN BIVALENT LOGIC

Embedding Logic and Semantics in Dynamics Systems

Rescher's V_{vp} Schema for Many-Valued Logic



John Tenniel classic drawing, "The Mad Hatter" from Lewis Carroll's *Alice in Wonderland*



Caricature of Bertrand Russell by Vicky, *New Statesman*, reprinted in Russell's *Wisdom of the West* (Crescent Books, 1959, p. 313).

- Assume $/p/$ = the actual value of 'Bertrand Russell looks like the Mad Hatter' = $2/3$
- How true is the statement: "It is half-true that Bertrand Russell looks like the Mad Hatter"?
- $.V_{vp}$ = the value of the proposition V that proposition p has the value v is true to the extent the p 's attributed value does not differ from its actual value.

$$V_{vp} = 1 - \text{ABS} (v - /p/) = 1 - \text{ABS} (1/2 - 2/3) = 5/6.$$

LOGISTIC LIAR

$$x_{n+1} = 1 - \text{ABS}((1 - x_n)^2 - x_n)$$

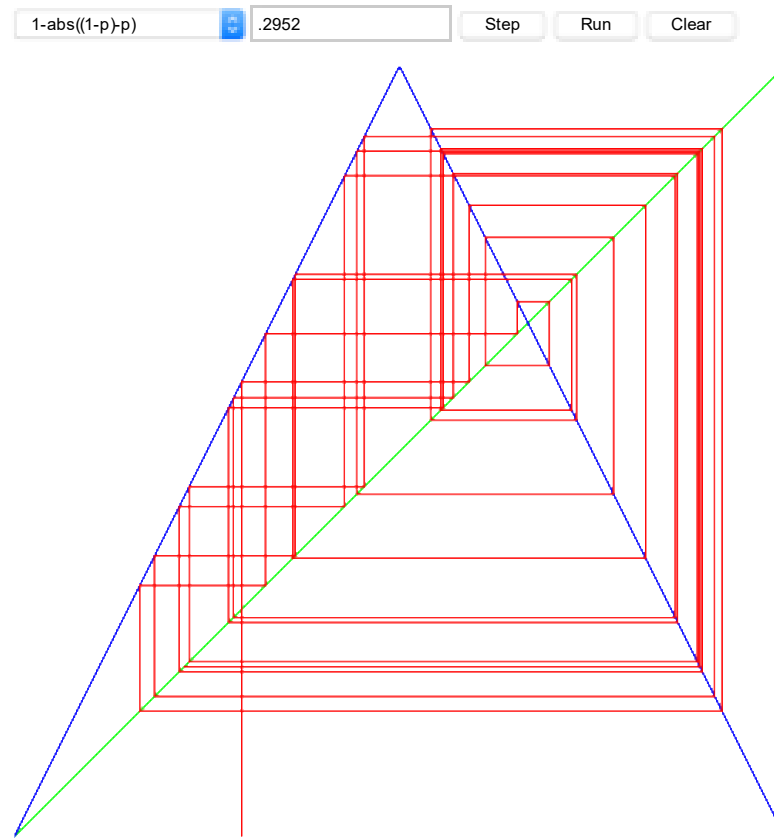


CHAOTIC LIAR

$$x_{n+1} = 1 - \text{ABS}((1 - x_n) - x_n)$$



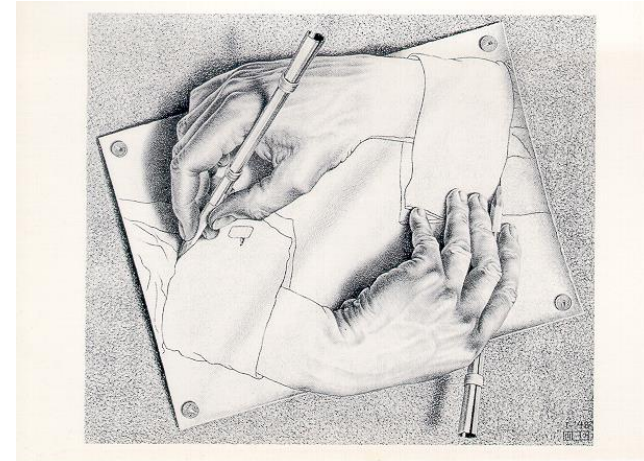
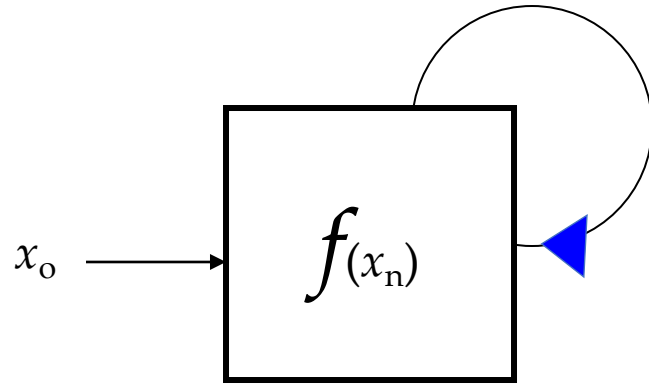
CLASSICAL LIAR



This sentence is as true
as it is estimated to be *false*.

Self-Reference as Iterated or updated evaluation

“The qualitative study of unstable aperiodic behavior in deterministic nonlinear dynamical systems.”



A New Menagerie of **Dynamic Quasi-Paradoxical** Sentences

$$x_{n+1} = 1 - \text{ABS} (\mathbf{S}(x_n) - x_n)$$

Model **self-reference** as functional feedback, dynamic or **iterative** updating

Replace a static **proposition** with a dynamic **saying** $\mathbf{S}(x_n)$

Replace the stable "**actual value**" of propositions with a **series** of **estimated values**.

- **CLASSICAL LIAR**: "*I am false*".

$$x_{n+1} = 1 - \text{ABS} (\mathbf{0} - x_n)$$

- **CONTINUOUS-VALUED LIARS**: "*I have the truth-value v* ".

- **CLASSICAL LIAR** for $v = 0$.
- Rescher's fixed-point "solution" for $v = 1/2$.
- Kripke's **TRUTH-TELLER** for $v = 1$.

- **CAUTIOUS TRUTH-TELLER**: "*I am **half** as true as I am estimated to be true*".

$$x_{n+1} = 1 - \text{ABS} (\mathbf{1/2} * x_n - x_n)$$

- **CONTRADICTIONARY LIAR**: "*I am as true as the **conjunction** of my estimated value and the estimated value of my **negation***".

$$x_{n+1} = 1 - \text{ABS} (\mathbf{MIN} \{x_n, \mathbf{1} - x_n\} - x_n)$$

Dualist/Escape-Time Diagram $\Rightarrow$ Fractals

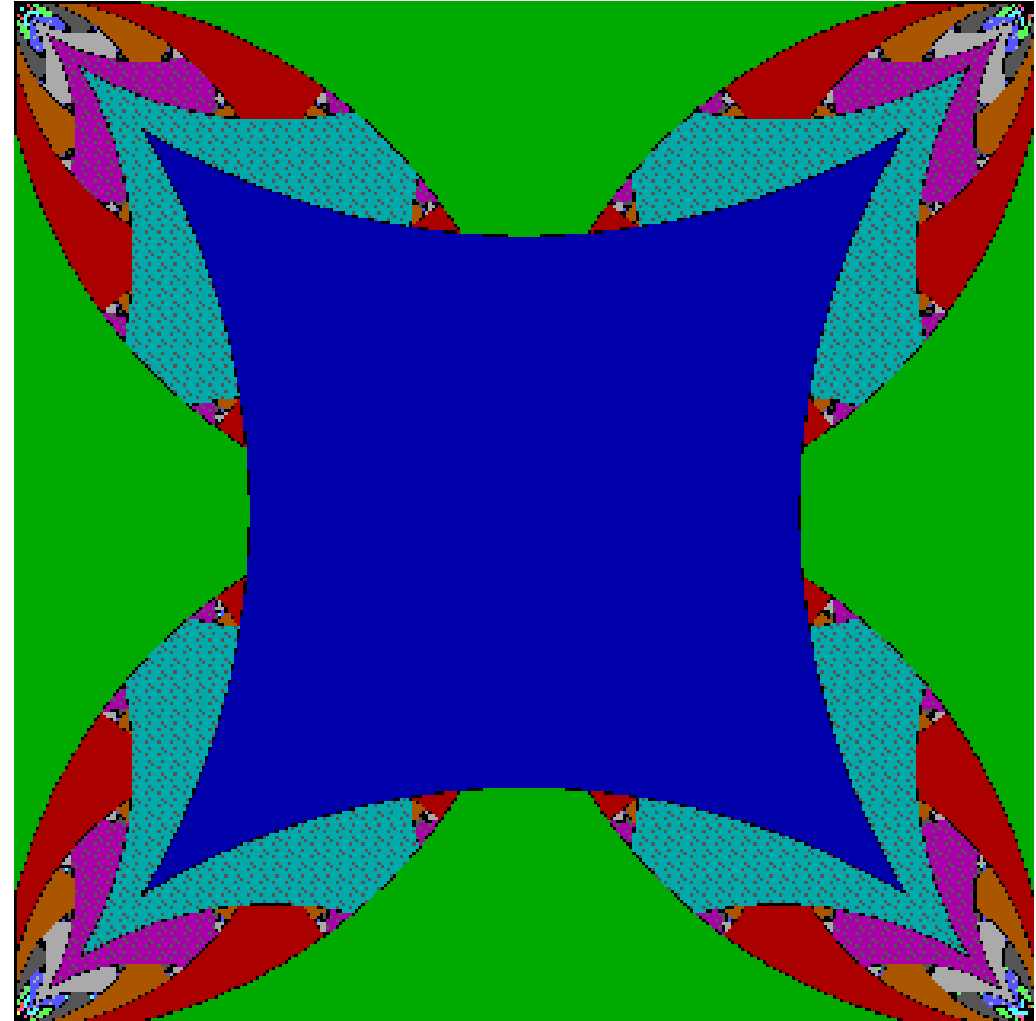
Aristotle, “What Plato says is *true*.”

Plato, “What Aristotle says is *false*.”

DUALIST LIAR: a *pair* of dynamical systems

$$x_{n+1} = 1 - \text{ABS } y_n$$

$$y_{n+1} = 1 - \text{ABS } [(1 - x_n) - y_n]$$



Gödelian Theorem (Rice's Theorem)

Either the boxed sentence is
semantically chaotic or it is as
true as it is estimated to be false.

Does the **STRENGTHENED CHAOTIC LIAR** have
chaotic semantic behavior or not?

*If it does, then the first disjunct is constantly
true and so the disjunction is not chaotic.
Contradiction.*

*If it does not have chaotic semantic behavior,
then the Strengthened Chaotic Liar is
equivalent to the second disjunct, which is
semantically chaotic. Contradiction.*

Non-Calculability of Chaos. Let C be the set of *chaotic functions* defined on the set of *partially recursive* functions on the real interval $[0,1]$. Given $\lambda x(1-\text{ABS}[(1-x)-x]) \in C$ and $\lambda x[1] \notin C$, the index set $I(C) = \{i: f_i \in C\}$ is not effectively calculable.

Assume $I(C)$ is λ -calculable. Then $\lambda x(1-\text{ABS}[(1-x)-x])$ is some $f_i \in C$ and $\lambda x[1]$ is some $f_j \notin C$. Then we may define the *diagonal function*

$$d(x) = \begin{cases} j & \text{if } f_x \in C \\ i & \text{otherwise.} \end{cases}$$

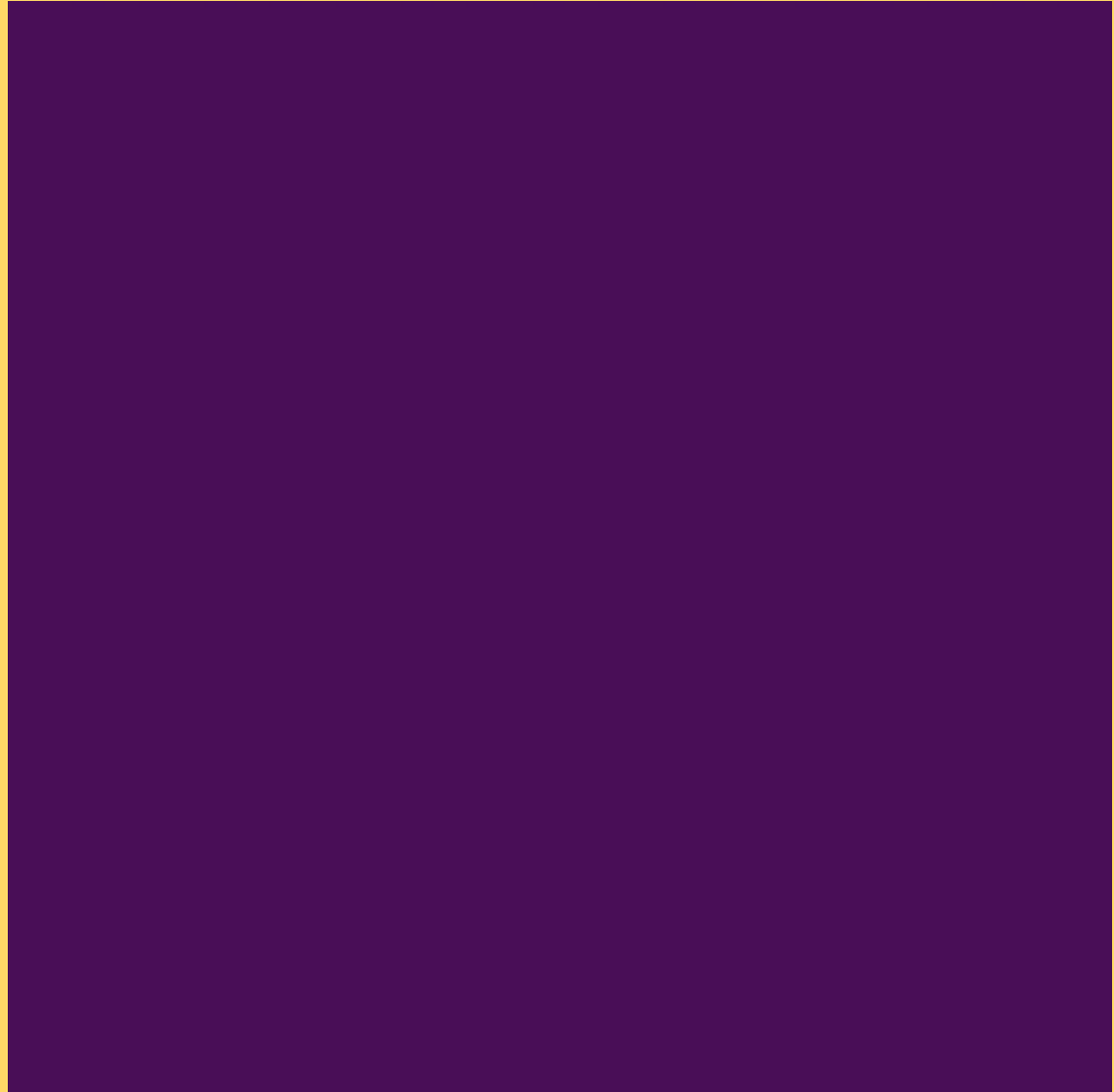
By the fixed-point lemma, there exists k , $f_k = f_{d(k)}$ so

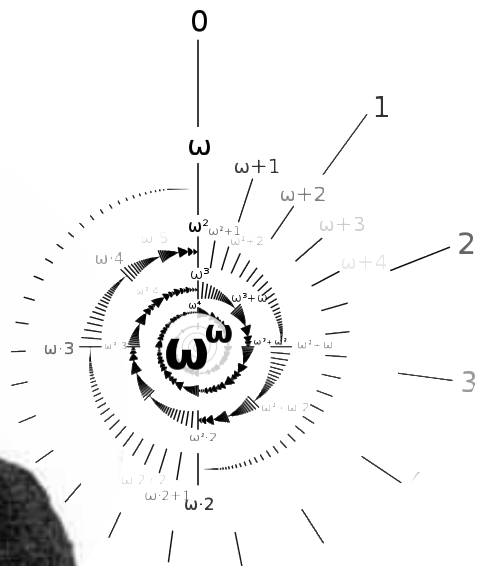
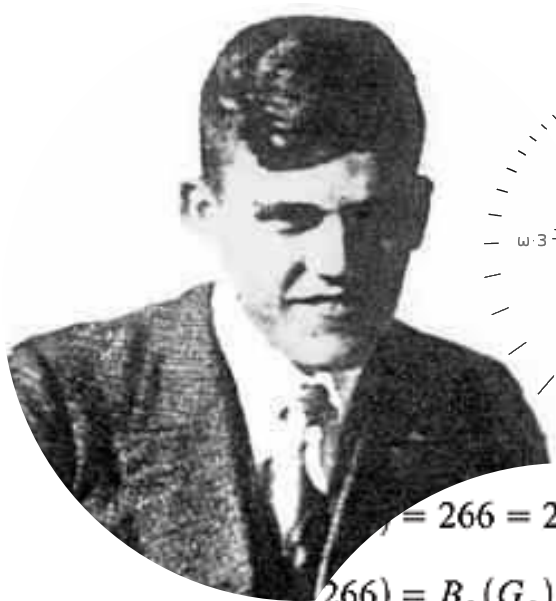
$$f_k \in C \text{ iff } d(k) = j .$$

But $f_j(x) = \lambda x[1]$, which is the non-chaotic constant function identical to 1. So we have $f_k \in C$ iff $f_k \notin C$.

Therefore, $I(C)$ is not λ -calculable and so not effectively calculable.

Escape-Time diagram for Dualist
Liars with constantly changing
parameter accompanied by AI
generated music.





$$\begin{aligned}
 266 &= 2^{2^{2+1}} + 2^{2^{2+1}} \\
 i_1(266) &= B_2(G_0) - 1 = 3^{3^{3+1}} + 3^{3^{3+1}} \\
 i_2(266) &= B_3(G_1) - 1 = 4^{4^{4+1}} + 4^{4^{4+1}} \\
 i_3(266) &= B_4(G_2) - 1 = 5^{5^{5+1}} + 5^{5^{5+1}} \\
 i_4(266) &= B_5(G_3) - 1 = 6^{6^{6+1}} + 6^{6^{6+1}} \\
 &= 6^{6^{6+1}} + 5 \cdot 6^6 + 5 \cdot 6^5 + \dots + 5 \\
 i_5(266) &= B_6(G_4) - 1 \\
 &= 7^{7^{7+1}} + 5 \cdot 7^7 + 5 \cdot 7^5
 \end{aligned}$$

Gödel criticized those who advocated for a mechanistic model of the mind failed to consider that minds are evolving over time.

... What Turing disregards completely is the fact the mind, in its use, is not static but constantly developing..." [1972, 306].

$$\varepsilon_0 = \omega^{\omega^{\omega^{\dots}}}.$$

Turing was aware that his ordinal construction attempted to overcome one specific type of Gödelian incompleteness, one that involved a relatively artificial construction of a true but unprovable sentence.

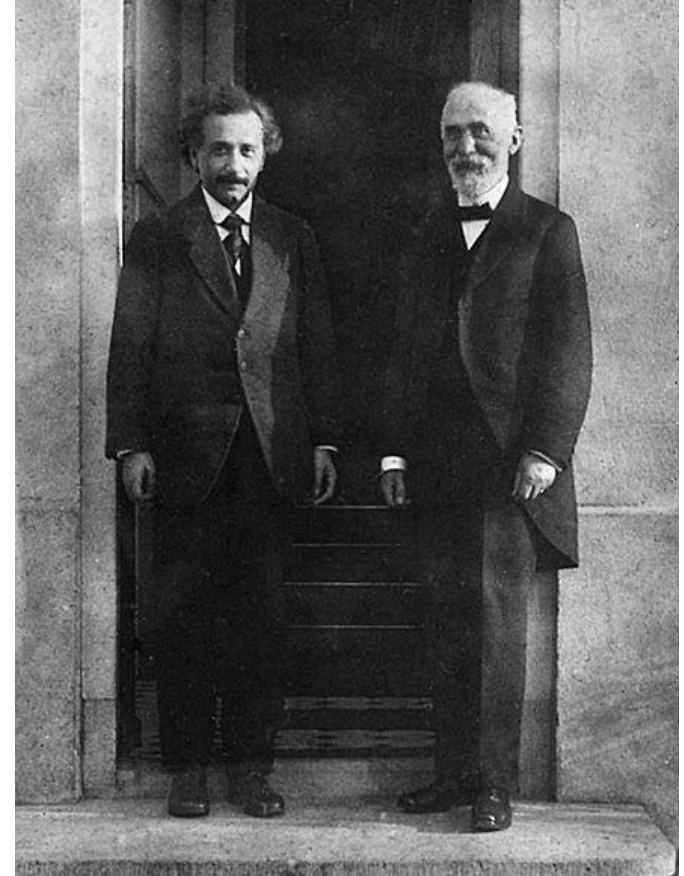
Gentzen's [1936] idea for overcoming Gödel's Second Incompleteness Theorem is analogous to Fermat's Disproof by Infinite Descent. Just as there can be no infinitely descending sequence of natural numbers in $\mathbb{N}$, there can be no infinitely descending chain of ordinals/

Einstein – Bergson Debate

In his debate with the French philosopher Henri Bergson on April 6, 1922, who claimed to be “more Einstein than Einstein” during his 20-minute critique, Einstein replied with one sentence:

“Il n’y a donc pas un temps des philosophes”

There was pay back. When Einstein was awarded the 1921 Nobel Prize in 1922, the Nobel committee invoked Bergson’s name, and awarded Einstein the prize, not his theory of relativity, but for his paper on the photoelectric effect. Einstein accepted the prize and talked about his theory of relativity.





"I'm sure I'll take you with pleasure!" the Queen said. "Two pence a week, and jam every other day."

Alice couldn't help laughing, as she said, "I don't want you to hire me – and I don't care for jam."

"It's very good jam," said the Queen.

"Well, I don't want any to-day, at any rate."

"You couldn't have it if you did want it," the Queen said. "The rule is, jam to-morrow and jam yesterday – but never jam to-day."

"It must come sometimes to 'jam to-day'," Alice objected.

"No, it can't," said the Queen. "It's jam every other day: to-day isn't any other day, you know."

"I don't understand you," said Alice. "It's dreadfully confusing!"

— Lewis Carroll, *Through the Looking Glass* [1871]

(1) $\blacktriangleleft \blacksquare \blacktriangleleft J$

(3) $\blacklozenge \blacksquare \neg J$

(5) $\blacksquare (J \rightarrow \blacktriangleleft \neg J)$

(7) $\blacksquare (J \vee \blacktriangleleft J)$

(2) $\square \triangleright \neg J$

(4) $\diamond (J \wedge \triangleright J)$

(6) $\square (J \rightarrow \triangleright \square \neg J)$

(8) $\square [(J \rightarrow \triangleright \neg J) \wedge (\neg J \rightarrow \triangleright J)]$

- (A) Alice will eat jam at most once.
- (B) Before a finite number of days ago, Alice had never eaten jam.
- (C) Until the day before yesterday, Alice always had eaten jam.
- (D) Alice has never eaten jam two (or more) days in a row.
- (E) There has never been two or more days in a row in which Alice did not eat jam.
- (F) At some time in the future, Alice will eat jam two or more days in a row.
- (G) Alice will never eat jam after today.
- (H) Alice will eat jam every other day.

Axioms for Time

$\Diamond_F T$	<i>future succession</i>	$\forall x \exists y \ x < y$
$\Diamond_P T$	<i>past succession</i>	$\forall x \exists y \ x > y$
$\Diamond_F \Diamond_F \varphi \rightarrow \Diamond_F \varphi$	<i>future transitivity</i>	$\forall xyz \ (x < y < z \rightarrow x < z)$
$\Diamond_P \Diamond_P \varphi \rightarrow \Diamond_P \varphi$	<i>past transitivity</i>	$\forall xyz \ (x > y > z \rightarrow x > z)$
$\Diamond_F \varphi \rightarrow \Diamond_F \Diamond_F \varphi$	<i>future density</i>	$\forall x > t \ \exists y > t : x > y$
$\Diamond_P \varphi \rightarrow \Diamond_P \Diamond_P \varphi$	<i>past density</i>	$\forall x < t \ \exists y < t : x < y$
$\varphi \wedge \Box_H \varphi \rightarrow \Diamond_F \Box_H \varphi$	<i>future discreteness</i>	$\forall t : \exists x > t \ \forall y < x \ (y = t \vee y < t)$
$\varphi \wedge \Box_G \varphi \rightarrow \Diamond_P \Box_G \varphi$	<i>past discreteness</i>	$\forall t : \exists x < t \ \forall y > x \ (y = t \vee y > t)$
$\Diamond_F \varphi \rightarrow \Box_G (\Diamond_F \varphi \vee \varphi \vee \Diamond_P \varphi)$	<i>future linearity</i>	$\forall t : \forall x > t \wedge \forall y > t \ (y < x \vee y = x \vee x < y)$
$\Diamond_P \varphi \rightarrow \Box_H (\Diamond_P \varphi \vee \varphi \vee \Diamond_F \varphi)$	<i>past linearity</i>	$\forall t : \forall x < t \wedge \forall y < t \ (y < x \vee y = x \vee x < y)$

Two higher-order properties include:

Continuity: every temporal subset with an upper bound has a least upper bound.

Homogeneity: every temporal point can be mapped onto any other by some order-isomorphism.

	AXIOMS	GÖDEL COMMENTS
	<u>Ax1</u> $P(\phi) \wedge P(\psi) \rightarrow P(\phi \wedge \psi)$	"and for any number of summand"
	<u>Ax2</u> $P(\phi) \vee \neg P(\phi)$	"exclusive or"
	<u>Ax[3]</u> $P(\phi) \rightarrow \Box P(\phi)$ $\neg P(\phi) \rightarrow \Box \neg P(\phi)$	"because it follows from the nature of the property" "But if a system S of pos. props. were incompatible would mean that the sum props. S (which is x) would be $x \neq x$."
GÖDEL'S AXIOMS	<u>Ax4</u> $P(\phi) \wedge (P(\phi) \Rightarrow \psi) \rightarrow P(\psi)$ which implies $x = x$ is positive $x \neq x$ is negative	"Positive means positive in the moral aesthetic independently of the accidental structure of the Only then the ax. true. It may also mean pure 'attri as opposed to 'privation' (or containing privatio interpret. Simpler proof." "If ϕ pos, then not: $(x)\Box \neg \phi(x) \rightarrow x \neq x$ hence $x \neq x$ pos: $x = x$ neg contrary Ax 4 on the exist. of pos. prop." "i.e., the 'disj' normal form in terms of elem. prop. [a] member without negation."
Scott's version of Ax2	$P(\neg \phi) \leftrightarrow \neg P(\phi)$ Separate into two conditionals: (A) $Pos(\phi) \rightarrow \neg Pos(\neg \phi)$ (B) $\neg Pos(\phi) \rightarrow Pos(\neg \phi)$	[Scott 1987] "that is either the property or its n positive, but not both." Anderson [1990]: Axiom (B) "seems to overlook a p that both a property and its negation should be n Only (A) is required for Gödel's proof; the question required for the proof that all truths are necessary.
Anderson's Emendation	Axiom 0. $Pos(\phi) \rightarrow \Box Pos(\phi)$ Axiom 1. $Pos(\phi) \rightarrow \neg Pos(\neg \phi)$	Anderson [1990]: "A property is positive if its abs entity entails that the entity is imperfect and its pre not entail [that]" [1990], p. 297.

Outline of Proof of GöD

Axiom 1. $\forall \phi [P(\phi) \leftrightarrow \sim P(\sim \phi)]$. "Either a property ϕ or its negation $\sim \phi$ is positive, but not both."

Axiom 2. $\forall \phi \forall \psi [P(\phi) \wedge \Box \forall x [\phi x \rightarrow \psi x] \rightarrow P(\psi)]$. "Any property entailed by a positive property is positive."

Theorem 1. $\forall \phi [P(\phi) \rightarrow \Diamond \exists x \phi x]$. "Any positive property is possibly exemplified."

Definition 1. $Gx \leftrightarrow \forall \phi [P(\phi) \rightarrow \phi x]$. "A god-like being possesses all positive properties."

Axiom 3. $P(G)$. "The property of being god-like is positive."

Axiom 4. $\forall \phi [P(\phi) \rightarrow \Box P(\phi)]$. "All positive properties are necessarily positive."

Definition 2. $\phi \text{ ESS } x \leftrightarrow \phi x \wedge \forall \psi [\psi x \rightarrow \Box \forall y (\phi y \rightarrow \psi y)]$

"An *essence* of an individual is a property it possesses that necessarily implies any other properties it possesses."

Theorem 2. $Gx \rightarrow G \text{ ESS } x$ "Being god-like is an essence of any god-like being."

Definition 3. $NEx \leftrightarrow \forall \phi (\phi \text{ ESS } x \rightarrow \Box \exists x \phi x)$.

"An individual has *necessary existence* iff all of its essences are necessarily exemplified."

Axiom 5. $P(NE)$ "Necessary existence is a positive property."

Theorem 3. $\Box \exists x Gx$ "It is necessary that God exists."



"The real argument for objectivism is the following. We know many general propositions about natural numbers to be true (2 plus 2 is 4, there are infinitely many prime numbers, etc.) and, for example, we believe that Goldbach's conjuncture makes sense, must be either true or false without there being any room for arbitrary convention. Hence, there must be objective facts about natural numbers. But these objective facts must refer to objects that are different from physical objects because, among other thing they are unchangeable in time."

The hope of finding "...a proof for freedom from contradiction undertaken by Hilbert and his disciples" has "vanished entirely in view of some recently discovered facts. It can be shown quite generally that there can exist no proof of the freedom of contradiction of a formal system *S* which *could be expressed in terms of the formal system S itself*"

"Time has no specific character of being. In relativity theory the temporal relation is like far and near in space. I do not believe in the objectivity of time. The concept of *Now* never occurs in science itself, and science is supposed to be concerned with all that is objective."

"My belief is theistic; not pantheistic, following Leibniz rather than Spinoza."

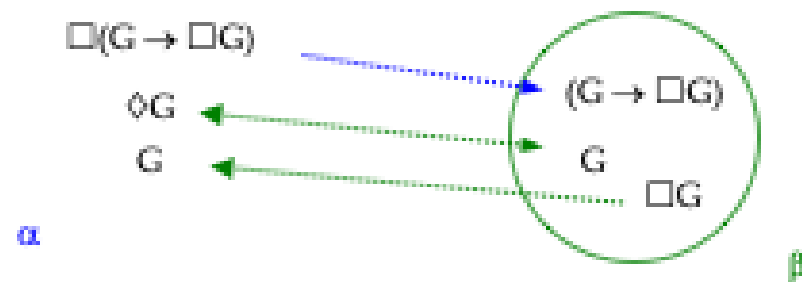
"Spinoza's God is less than a person. Mine is more than a person.... "

"Now he has departed from this strange world a little ahead of me. That signifies nothing. For those of us who believe in physics, the distinction between past, present and future is only a stubbornly persistent illusion."

"I go to my office at the Institute for Advanced Study just to have the privilege of walking home with Kurt Gödel."

"I believe in Spinoza's God, Who reveals Himself in the lawful harmony of the world, not in a God Who concerns Himself with the fate and the doings of mankind."

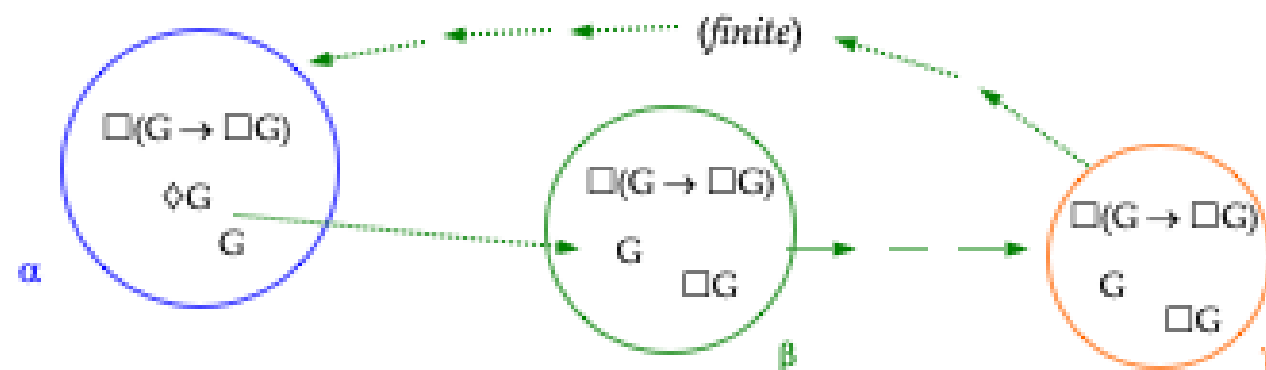
"How can it be that mathematics, being, after all, a product of human thought which is independent of experience, is so admirably appropriate to the objects of reality? Is human reason, then, without experience, merely by taking thought, able to fathom the properties of real things?"



In fact, the Brouwersche Axiom allows the deduction of the *stronger* conclusion $\Box G$, namely that God's existence is *necessary*. A *weaker* condition suffices for a proof of God's *actual* existence:

$$\Diamond G \wedge \Box(G \rightarrow \Box G) \rightarrow G$$

namely, the principle of "finite return"¹, i.e., from each successor world we can always return to α by a finite chain of successor worlds:



¹ L. G. E. Brouwer, *Philosophy* 1973, 174.

The significance of mathematical logic for philosophy lies in its power to make thoughts explicit by illustrating the providing a frame for the axiomatic method. Mathematical logic makes explicit the central place of predication in the philosophical foundation of rational thought.¹

The fundamental principles are concerned with what the primitive concepts are and also their relationship. The axiomatic method goes step by step. We continue to discover new axioms; the process never finishes. Leibniz used formal analogy: in analogy with the seven stars in the Great Bear constellations, there are seven concepts. One should extend the analogy to cover the fact that by using the telescope we [now] see more stars in the constellation.²



Like the North Star in Ursa Major was used for navigation for centuries, Gödel's theorems have been a guiding light for philosophical exploration: he produced elegant theorems not only in mathematical logic, but also results approaching the status of theorems about the creativity of the mind and mechanistic calculation, about the modal status time in relativity physics, and a deduction of a God with all perfections (or positive properties) in theology. The seven points in Gödel's constellation explicated in this essay are:

1. "The world is rational."
2. "There is incomparably more knowable *a priori* than is currently known."
3. If (1) and (2), then logic, mathematics, and physics reveal the rational structure of the world.
4. Mechanistic formal systems cannot be the cause of mathematical rationality because mathematical logic can be used to prove the limitations of such formal systems.
5. All mathematical truth cannot be captured by mechanistic formal systems (a philosophical implication of the Incompleteness Theorems).
6. If Einstein's relativistic physics reveals the nature of physical reality, then (physical) time is not objectively real.
7. The only sufficient reason, or First Cause, of mathematics, logic, and physics is the mind of God; hence, there necessarily exists a unique, god-like being.

¹ Wang [1997], 293, 9.1.16.

² Wang [1996], Chapter 9, "Gödel's Approach to Philosophy", 9.5.6, 320.

REPLY TO OBJ. 2. The diverse purposes for proofs



Gödel's Letter to Burke Grandjean dated 19 August 1975: "Replying to your inquiries I would like to say first **that I *don't* consider my work as a "facet of the intellectual atmosphere of the early 20th century," but rather the opposite.** It is true that my interest in the foundations of mathematics were aroused by the "Vienna Circle," but the philosophical consequences of my results, as well as the heuristic principles leading to them, are anything but positivistic or empiricistic." [GCW-IV, p. 443]



Wittgenstein: A proof of God's existence ought really to be something by means of which one could convince oneself that God exists. **But I think that what *believers* who have furnished such proofs have wanted to do is to give their 'belief' an intellectual analysis and foundation, although they themselves would never have come to believe as a result of such proofs.** Perhaps one 'convince someone that God exists' as a result of such proofs. Perhaps one could convince someone that God exists' by means of a certain kind of upbringing, by shaping his life in such and such a way. (quoted in Wang [1996], 113)